

AI Images, Trust and News Demand

Maja Adena

WZB Berlin

Eleonora Alabrese

University of Bath

Francesco Capozza

University of Barcelona

Isabelle Leader

University of Bath

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Abstract

We test whether AI-generated news images affect demand and trust of outlets. In two pre-registered experiments with UK adults, we vary: image origin (wire-service photograph vs. matched AI image), source labels and AI photorealism, and measure newsletter sign-up as a behavioral proxy for news demand, alongside perceived outlet quality. The average demand is unaffected by AI imagery or labels. This aggregate null conceals sharp belief-driven heterogeneity: audiences who recognize an image as AI-generated reduce engagement and downgrade outlet's quality by approximately 10 percentage points, while those who do not show the opposite response. Labels do little to correct mistaken attributions. Instrumenting for recognition with randomized photorealism, we find a precise null effect on newsletter demand and a 0.35 standard deviation decline in perceived outlet quality. A complementary survey of 132 international journalists finds that practitioners overestimate baseline demand and anticipate large negative effects of AI imagery, which are absent in the data. What matters is not whether an image is AI-generated, but whether audiences believe it is.

Keywords: AI; Demand for News; Trust; Online Experiment.

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Adena: WZB, TU Berlin, CESifo, maja.adena@wzb.eu; **Alabrese:** University of Bath, CAGE, SAFE, ea2143@bath.ac.uk; **Capozza:** University of Barcelona, IEB, CESifo, francesco.capozza@ub.edu; **Leader:** University of Bath ial35@bath.ac.uk. IRB was obtained by the University of Bath with the number 10699-12847. The experiment 1 has been pre-registered on AsPredicted with the number [251218](#), while the second experiment has been pre-registered on OSF with the number [e5ncv](#). The authors report no conflict of interest. The authors declare no competing interests. M.A., E.A., F.C., and I.L. designed research; performed research; analyzed data; and wrote the paper.

Significance Statement

As newsrooms increasingly adopt AI-generated imagery, a critical question emerges: does this erode public trust and engagement? Using two pre-registered experiments with over 7,000 UK adults, we show that AI images do not reduce newsletter demand on average, but this aggregate null conceals sharp, belief-driven heterogeneity. Audiences who recognize an image as AI-generated reduce engagement and downgrade outlet's quality by roughly 10 percentage points; those who do not recognize it, show the opposite response. AI labels only marginally correct mistaken attributions. A causal test using randomized AI photorealism confirms that recognition (not mere exposure) drives losses in outlet's standing by 0.35 standard deviations. Journalists systematically overestimate these demand penalties. What matters is not whether an image is AI-generated, but whether audiences believe it is.

1 Introduction

Public trust in news has been low and declining in many countries since 2015, including the UK ([Reuters Institute for the Study of Journalism, 2024](#)). When audiences distrust the press, they discount journalistic information and rely more on partisan cues ([Ladd, 2010](#)). At the same time, synthetic media — i.e., AI-generated or manipulated content — is already entering political communication: campaign actors have begun using AI-generated visuals in advertising ([Novak, 2023](#)). Major platforms have responded by rolling out AI-content labels, including Meta and TikTok ([Nellis, 2024](#); [Paul, 2024](#)), aiming to increase transparency around AI-generated content. Recent events suggest, however, that disclosure is operationally fragile: in February 2026, it became known that Germany's public broadcaster ZDF aired AI-generated images in its *journal* without a corresponding on-screen notice; ZDF subsequently apologized, removed the segment across platforms, and stated that the material should not have been broadcast without clear labeling and editorial context.¹

Yet generative AI may further influence trust. Photorealistic synthetic faces are increasingly indistinguishable from real ones, and are sometimes judged as more trustworthy, raising concerns for news visuals ([Nightingale and Farid, 2022](#)). This led to growing interest in whether disclosure can mitigate misperceptions. Early evidence shows that labels carry meaning for audiences: people penalize headlines labeled as "AI-generated," even when the content is accurate or produced by humans ([Altay and Gilardi, 2024](#)), and AI-generated visuals can shift perceptions of political news ([Ash et al., 2021](#); [Caprini, 2024](#); [Park et al., 2024](#)). However, the effects of AI on trust remain uncertain, with ongoing debates about how to signal AI content, how audiences interpret these labels, and whether they increase or erode outlets' trust ([Wittenberg et al., 2025](#)).

¹*Die Zeit*, "ZDF entschuldigt sich für nicht gekennzeichnete KI-Bilder im Programm," updated February 17, 2026; and ZDF press release, "ZDF entschuldigt sich für Fehler im 'heute journal'," February 17, 2026.

While news outlets have long used image post-production without disclosing the underlying technology, generative AI is increasingly used to edit or even create entire visuals.² Large-scale audits likewise show that AI systems already support routine news production, often without disclosure (Matich et al., 2025; Russell et al., 2025). Audiences are aware of this shift: many suspect AI involvement yet remain unsure where it occurs, and they express more discomfort with visible generative use than with behind-the-scenes applications (Fletcher and Nielsen, 2024; Reuters Institute for the Study of Journalism, 2024). In this context, an AI-generated image (especially when labeled) may invite inferences on automation, cost-cutting, and reduced editorial oversight, carrying reputational costs that traditional editing does not.

We study whether AI-generated images used by news outlets, and their transparent reporting through labels, affect trust in and engagement with those outlets. In a pre-registered online experiment with 2,870 UK respondents, participants read the same article sourced from *The Conversation* paired with one of four visuals: a wire-service photo (with or without credit) or a matched AI image (with or without a AI label). The AI and wire-service images are nearly indistinguishable and depict the same scene. Participants know ex-ante that they may later subscribe to the outlet’s weekly newsletter. We measure (i) behavioural engagement (newsletter sign-up), and whether it varies with (ii) recognition that the image is AI-generated, and (iii) perceived outlet quality as a potential mechanism. This design isolates the causal effect of AI visuals and disclosure labels on outlet perceptions.

We posit three hypotheses. First, relative to the main control (a labelled wire-service photo), newsletter sign-up will be lower in the AI-image conditions (because attribution signals undermine trust). Second, the unlabelled AI image should have a weaker effect on news demand, since more participants might not detect its synthetic origin. Third, lower perceived outlet quality should mediate reduced engagement.

We first confirm that the treatment affects AI recognition, though it is generally poor. Even for the authentic wire-service photo, respondents often reported the image as AI-generated: 47.2% in the unlabelled control and 42.9% in the labelled control. Thus, adding a source label only modestly reduces mistaken AI attributions. Recognition increases significantly only when the image is AI-generated and labelled as such. These results highlight substantial baseline confusion: authentic photographs are often mistaken for AI, and labels improve recognition only when the image is synthetic, consistent with concerns about the difficulty of distinguishing real and synthetic visuals (Nightingale and Farid, 2022).³

On behaviour, we find a precise average null. Baseline newsletter demand in the unlabelled control is 36.5%, and the other treatment arms fall in a narrow band between 37.0 and 39.3%. Neither replacing the original photo with an AI image nor adding a transparent label meaningfully

²See [ADM+S Centre overview](#) for documented cases.

³The average dwell time on the article page was more than one and a half minutes with no differences by treatment. We therefore find attention reasonably high and balanced.

alters sign-ups.⁴ Taken at face value, these results indicate that neither images nor labels are a first-order lever on short-run engagement — directly informing the labeling policies rolled out by major platforms (Nellis, 2024; Paul, 2024) and debates about what users infer from AI disclosures (Altay and Gilardi, 2024; Wittenberg et al., 2025).

Behind this aggregate null, however, behaviour differs sharply depending on whether respondents correctly identified the image as AI. When we split the sample by recognition, those who mistakenly think AI pictures are real have 10 percentage points higher signups than those who mistakenly think a real picture is AI-generated. Similarly, those who correctly recognise AI pictures have newsletter signup rates about 10% lower than those who correctly recognise real pictures. AI boosts engagement when it goes unnoticed, but penalises the outlet when recognised. Labels modestly temper this penalty at the margin, without closing the recognition-based gap.

Taken together, the results suggest that what matters is not whether an image is AI, but whether audiences *believe* it is, and current source disclosures do little to correct mistaken attributions. This pattern is consistent with evidence that AI visuals can shift consumption by altering perceptions of the information environment (Campante et al., 2025) especially in political-news settings (Caprini, 2024; Park et al., 2024), while disclosure effects remain mixed (Altay and Gilardi, 2024; Wittenberg et al., 2025). It also parallels research on algorithm aversion even when algorithmic performance is strong (Dietvorst et al., 2015; Dominguez-Castelo et al., 2019).

From the outlet’s perspective, these responses matter because they arise even when AI delivers high-quality visuals. Industry surveys report strong enthusiasm for generative tools to reduce repetitive work and expand visual formats (Beckett and Yaseen, 2023; Diakopoulos et al., 2024). Our results show that audiences impose a reputational penalty once AI use is recognised, even though the underlying quality is unchanged.

Beliefs about the outlet move in tandem with behaviour. Among respondents who mistakenly classify AI images as real, perceived outlet quality is about 0.50–0.60 standard deviations higher than among those who mistakenly classify real images as AI-generated. The pattern reverses among respondents who correctly classify the images: perceived outlet quality is about 0.45–0.55 standard deviations lower for those who correctly identify AI images than for those who correctly identify real images. These results are consistent with models linking credibility beliefs to news demand, and evidence using newsletter sign-ups as a behavioural proxy (Chopra et al., 2024). In other words, audiences punish outlets when they *believe* an image is synthetic — even when the image is real — and current labelling practices do little to prevent this, a mechanism with direct relevance for outlets and platforms navigating AI use and disclosure (Nellis, 2024; Paul, 2024; Reuters Institute for the Study of Journalism, 2024).

To interpret these patterns, we develop a simple Bayesian framework in which readers update

⁴The coefficients on AI, Label, and their interaction are all statistically indistinguishable from zero.

beliefs about outlet quality when they think the outlet uses AI. The model clarifies when AI can increase demand (if undetected) and when it reduces demand (if recognised), delivering predictions that match the empirical results. These dynamics are consistent with classic models of markets where quality is hard to observe and reputational signals drive behaviour (Klein and Leffler, 1981; Shapiro, 1983). Media outlets invest in credibility for this reason (Gentzkow and Shapiro, 2006; Gentzkow et al., 2025). In our setting, recognised AI use acts as a negative signal about the outlet’s type — lowering perceived accuracy, trustworthiness, and engagement — even though the content itself is unchanged. For outlets, the reputational cost of being perceived as "the kind that uses AI" may outweigh the productivity gains that generative tools promise.

We then address two natural concerns about Experiment 1: (i) that results may be specific to a single article and (ii) that the belief-based heterogeneity we document is not causally identified. We fielded a second pre-registered experiment with 4,200 UK respondents. Experiment 2 preserves the core origin×label design while extending it in two directions.

First, we replace the single article–image pair with a set of six news articles drawn from diverse topic areas (Business & Economy, Politics & Society, Arts & Culture, Education and Environment), all sourced again from *The Conversation*. This multi-stimulus design allows us to assess robustness of our main findings across different contexts. Second, and more importantly, we introduce randomised variation in the photorealism of the AI images. Each article is either paired with its original wire photograph or two matched AI-generated versions, created via a two-step protocol that separates scene description from image generation: either a highly photorealistic version (*very realistic AI*) or an intentionally less photorealistic version (*not very realistic AI*), while the depicted scene and semantic content are held constant. This manipulation exogenously shifts the probability that respondents recognize the image as AI-generated, without altering image origin or labeling. Because this manipulation is exogenous, we can use treatment assignment as an instrument to isolate the causal role of recognition — the channel our Experiment 1 results implicate — rather than simply conditioning on endogenous post-treatment beliefs.

Experiment 2 replicates both findings across six articles and twelve AI-generated images. Newsletter demand remains statistically indistinguishable from control in every treatment arm, while the sign reversal by recognition group — unrecognised AI raising demand, recognised AI reducing it — replicates with comparable magnitude. The photorealism manipulation successfully shifts recognition: respondents assigned the less realistic image are substantially more likely to identify it as synthetic (81.2% versus 44.1% in unlabelled conditions, $p < 0.001$). Instrumenting for recognition with this exogenous variation, we find that a one-unit increase in the probability of correct recognition leaves newsletter demand unchanged (2SLS: +0.015, s.e. 0.050, $p > 0.10$) but lowers perceived quality by 0.348 standard deviations ($p < 0.001$). These IV estimates align in sign and magnitude with the OLS heterogeneity from Experiment 1, supporting a causal interpretation of the belief-driven mechanism.

To assess external validity and practitioner beliefs, we complement the two experiments with a survey of 132 international journalists recruited via Prolific (see [Berger et al., 2026](#), as an example of studying journalists). We elicit their beliefs about how AI-generated images affect news demand (anchored to the baseline newsletter sign-up rate from Experiment 2). Journalists’ beliefs diverge from the data in both directions. They substantially overestimate baseline engagement and anticipate a large demand penalty from AI recognition — a penalty our experimental data do not support. Moreover, while they almost unanimously support AI disclosure, they underestimate peer support for it and overestimate how widely AI is already used in the profession. Together, these patterns suggest that the reputational risks of AI visual adoption are not fully reflected in current newsroom decision-making.

Contribution to the literature. These findings speak directly to how AI reshapes trust and news demand in realistic media environments. Our paper contributes to three strands of research.

Trust in media. We connect to economic models in which trust is endogenous and shapes information acquisition and inference ([Gentzkow et al., 2025](#)), and to evidence on how news outlets affect accountability and civic life ([Snyder and Strömberg, 2010](#)). We provide causal evidence whether the *use of AI-generated images* and their transparent disclosure affect perceived outlet quality and downstream engagement. This complements recent field evidence on AI, trust, and consumption with major news outlets ([Campante et al., 2025](#)). Whereas their work shows that making AI-generated misinformation salient can increase engagement by heightening concern about the information environment, we isolate a distinct mechanism: holding text, topic, and outlet constant, *the visual origin of a single image* can alter trust and behaviour, and penalties arise when audiences *believe* an image is synthetic, even when it is authentic.

News demand. We extend experimental work measuring newsletter demand and the role of accuracy concerns versus belief-confirmation motives ([Chopra et al., 2022, 2024](#); [Faia et al., 2024](#)), as well as research on fact-checking and belief updating ([Barrera et al., 2020](#); [Henry et al., 2022](#)). We show that the *visual format* and *source labels* causally shape beliefs about outlet quality and actual engagement. Specifically, audiences reduce outlet demand when they think a photo is synthetic (even when it is real) and increase it when they do not. Labels do little to correct mistaken attributions, likely because many viewers fail to attend to them. This isolates how the perceived visual credibility, rather than the text or topic, drives news demand within the same controlled article.

AI and the news ecosystem. We contribute to work on generative AI, platform responses, and the credibility of AI-generated content ([Altay and Gilardi, 2024](#); [Caprini, 2024](#); [Park et al., 2024](#); [Wittenberg et al., 2025](#)). Existing studies typically vary text, misinformation exposure, or platform-level policies. Our design instead isolates *source-level* consequences of embedding an AI image within a single controlled article, holding topic, text, and outlet constant. We also complement recent field evidence ([Campante et al., 2025](#)) by showing that penalties are

belief-driven and can arise even when the image is authentic, while labels do little to prevent mistaken inference. Finally, we relate to emerging work on AI-enabled news customization and its effects on matching and engagement (Chopra et al., 2025). Together, the experiment offers direct causal evidence on how AI visuals shape trust and outlets demand in a realistic media environment.

In the remainder of the paper Section 2 describes the experiment 1, Section 3 reports results from experiment 1. Section 4 describes the experiment 2, Section 5 reports results from experiment 2. Section 6 presents the survey we conducted with journalists. Section 7 discusses implications and concludes.

2 Experiment 1: Design and Sample

We run a controlled experiment in which respondents read the same news article, varying only the accompanying image and its source label. We begin by describing the sample and recruitment before outlining the detailed design.

2.1 Sample

Participants were recruited from the UK general population on Prolific.⁵ Before random assignment, respondents reported basic demographics (age, gender, education, region, employment, ethnicity, political affiliation, and income), allowing us to implement quotas on age, gender, and political affiliation to approximate the UK adult population. The survey also recorded news-consumption habits and familiarity with AI.

To ensure data quality, respondents completed a CAPTCHA, an attention check, and a short screening question on the use of large language models; anyone failing these checks was excluded, as pre-registered. After these exclusions, the final sample consists of 2,870 respondents (target $N \approx 3,000$). Table B.1 compares observable sample characteristics with UK population benchmarks: the sample is broadly similar, though somewhat more educated and having a higher income on average.

The study was pre-registered on AsPredicted (n. 251218).⁶ Prolific’s new built-in speed check feature was enabled to automatically exclude respondents completing the survey unreasonably fast relative to the median, ensuring sufficient time-on-task for all included participants. Consistent with this, the median completion time among the final sample was eight minutes. Respondents were paid £1.5 (over £11 per hour), in line with Prolific guidelines.

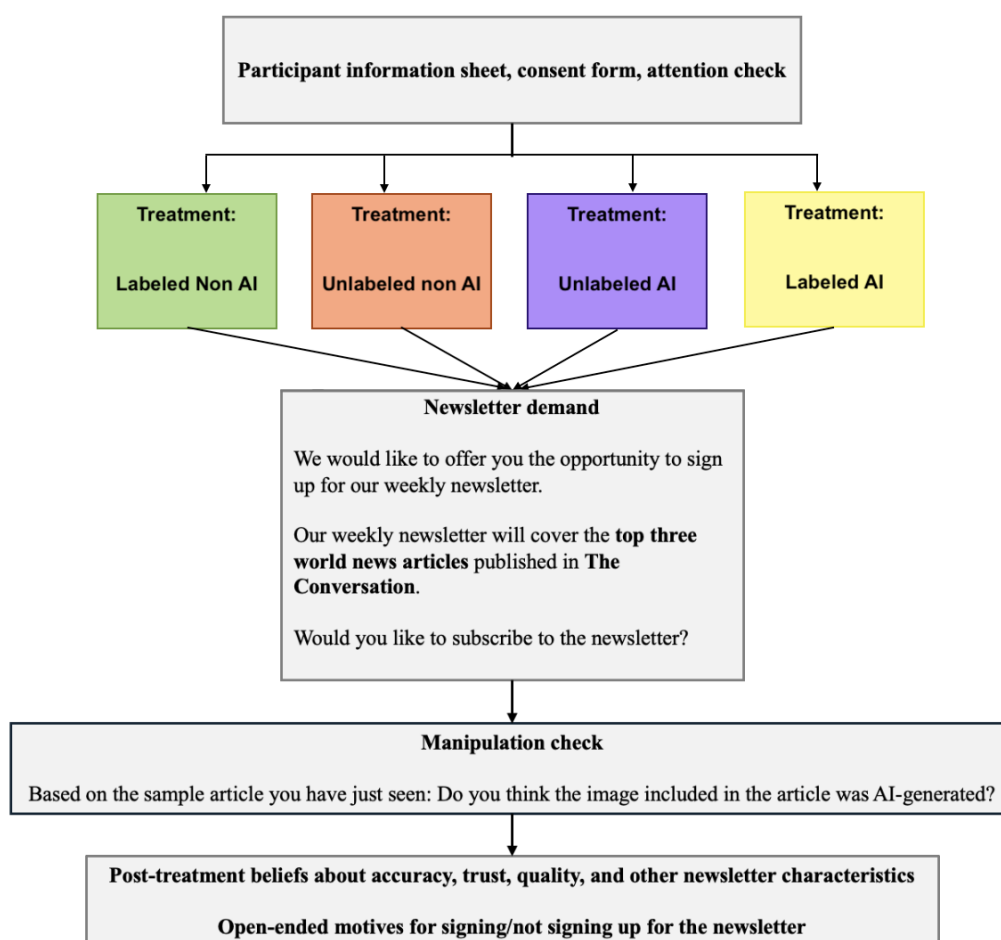
⁵Responses were collected between October 14 and 19, 2025.

⁶Pre-analysis plan available at: aspredicted.org/xw49-bcwX.pdf.

2.2 Experimental Design

After informed consent, standard quality checks, demographics, and media diet questions, each participant reads the same article. The article is accompanied by an image. Respondents are randomly assigned to one of four image conditions that vary only in (i) whether the accompanying picture is the original wire-service photograph or a closely matched AI-generated version, and (ii) whether a source label is displayed. Figure 1 provides an overview of the experiment, while Appendix E offers the complete set of instructions.

Figure 1: Experiment Overview



As an experimental stimulus, we use a real news article from *The Conversation*, a reputable UK academic-led outlet. The source of the news article is made salient at the outset such that subsequent evaluations are clearly attributed to the outlet. Importantly, while reading the article, our participants are aware of the subsequent possibility to subscribe to the weekly newsletter of the same outlet. The article discusses recent protests in California.⁷ For a UK sample, the topic is politically relevant yet psychologically distant, encouraging engagement while limiting strong

⁷Original article available at [this link](#).

personal identification or sacred-value reactions. Conflict between state and federal authorities also mirrors common themes in contemporary news, with competing narratives and potential for politically motivated misinformation, making it a realistic setting for studying credibility signals.

Treatment manipulation. The experimental manipulation concerns the accompanying image and its source label. In the two control groups, we display the article’s original photograph from EPA Images, a major photojournalism agency (Figure 2a). Importantly, the picture carries substantive informational content rather than serving a decorative role, so it plausibly shapes impressions of the outlet. In the two treatment groups, we replace the original photograph with a closely matched AI-generated version created using DALL-E (Figure 2b). Composition, lighting, and subject placement were designed to closely mirror the original so that visual content is comparable aside from image origin and labeling. The full text prompt appears in Appendix D.

Each image appears either with or without a source label. In the **Labeled Control**, the original credit ("*EPA-EFE/Caroline Brehman*") is displayed beneath the photo, as in the real article. In the **Labeled AI** condition, we show: "*AI-generated image (created using OpenAI’s DALL-E)*," matching The Conversation’s house style and current transparency expectations for AI disclosures (AP, 2023). **Unlabeled** conditions omit any credit or description.

Figure 2: Treatment Manipulation



(a) Original (Control with/without credit) **(b)** AI Image (Treatment with/without label)

To avoid deception, respondents are truthfully told that the text is from a real article. Because the possible use of AI images is not mentioned ex-ante, participants are debriefed at the end of the survey and provided with full information about synthetic picture and the objective of the study. We judged this omission necessary to preserve naturalistic responses to the article: explicitly warning participants would have primed them to scrutinize the image and undermined the treatment. The content was benign and posed minimal risk. Finally, Table B.2 shows that observable characteristics are balanced across the four experimental conditions.

A natural concern is that we rely on a single picture. However, the intervention does not

depend on the specific visual content of the image: it varies whether the image is original or AI-generated, and — crucially — whether its origin is disclosed. The AI and original images depict the same scene, and the only systematic difference across conditions is the presence or absence of a label. Prior evidence shows that AI labels reduce trust and engagement even when the underlying content is accurate or human-made (Altay and Gilardi, 2024), suggesting that responses generalise beyond any specific stimulus. Field evidence similarly indicates that making AI-generated content salient can shift news consumption by altering perceptions of the information environment, a mechanism that operates independently of the specific image shown (Campante et al., 2025). Moreover, humans struggle to distinguish synthetic from real visuals across diverse image sets, and photorealistic synthetic faces can even be judged as more trustworthy than real ones (Nightingale and Farid, 2022). Consistent with this broader literature, our effects reflect audience updating on the signal that an outlet uses AI, rather than on the content of any specific image.

Demand for news (primary outcome). Our primary behavioural outcome is newsletter subscription (yes/no). Prior experimental work validates newsletter sign-up as a revealed-preference proxy for news demand, showing how sign-ups respond to outlet credibility signals (e.g., fact-checking or outlet-bias cues). We follow this convention and treat newsletter sign-up as a compact, incentive-compatible engagement measure (Chopra et al., 2022, 2024). After reading the article coupled with its assigned image, participants choose whether they would like to receive a weekly newsletter summarizing the top three world news stories from *The Conversation*. Respondents who opt-in indicate interest in receiving a link via Prolific messaging system, without requiring participants to provide personal contact information. While the newsletter is free to access, subscribing still entails a non-monetary opportunity cost.

Manipulation check and recognition coding. After the primary outcome, we measure whether respondents correctly inferred how the image was created. We ask: “*Based on the sample article you just saw, was the image AI-generated?*” (Yes/No; forced choice, with no “Unsure” option). We then construct a recognition indicator:⁸

$$OriginCorrect = \begin{cases} 1 & \text{if respondent correctly identifies the image's origin (AI} \rightarrow \text{Yes; Real} \rightarrow \text{No)} \\ 0 & \text{otherwise.} \end{cases}$$

⁸For completeness, we also use the following indicator:

$$Manipulation = \begin{cases} 1 & \text{if respondent believes the image is AI-generated (Yes)} \\ 0 & \text{otherwise (No).} \end{cases}$$

Mechanisms (secondary outcomes). After the subscription decision and manipulation check, participants rate the outlet’s expected quality on three items: accuracy, trustworthiness, and overall quality (1–5 Likert scales, higher = better). We report results both item-by-item and using a standardized composite quality index (average of the three z-scored items). The index provides a proxy for perceived credibility, which we use to examine whether changes in engagement operate through belief updating about the outlet. In this index, positive values indicate above-average perceived quality and negative values indicate below-average perceived quality, allowing changes to be compared on a common scale.

Additional measures and demand checks. To assess alternative mechanisms and possible experimenter-demand effects, respondents also rated perceived political bias, complexity, entertainment value of the article, and trust in the researchers. We additionally included an open-ended question asking participants to guess the purpose of the study (Table B.9). A large share mentioned AI in general terms (53.5%), but almost none referred to the idea that AI images might affect trust in the outlet or willingness to subscribe. Most answers concerned broad themes such as “fake news,” “the rise of AI,” or “media bias,” rather than the treatment itself. This suggests that while AI was salient as a topic, participants did not infer our key behavioural hypothesis, and demand effects are unlikely to drive the results.

3 Results from Experiment 1

Across conditions, AI recognition is poor, and both outlet perceptions and demand behaviour hinge on whether people *believe* the image is AI-generated. Labels only marginally reduce mistaken attributions and do not eliminate the divide between those who think the image is AI and those who do not. We expand on these patterns below.

3.1 Manipulation: Poor AI recognition and weak labeling effects

Figure 3a reports positive responses to the question “*Was the image AI-generated?*” (*Manipulation=1*). With the authentic photograph and no credit, 47.2% of respondents said the image was AI. Adding the wire-service credit lowers this only slightly (to 42.9%). Showing an unlabeled AI image produces a similar response (45.8%). The only clear separation appears when the image is both AI *and* labeled: 51.6% of respondents say it was AI. The corresponding regression (Table B.3) confirms that the AI $\times$ Label interaction is positive and statistically significant (≈ 0.10 , $p < 0.01$). Even then, only about half of respondents identify the AI image as AI.

Figure 3b reports the share of correct responses to the same question (*OriginCorrect=1*), indicating whether a respondent correctly identified the image they personally saw as AI-generated or not. The only condition that clearly improves accuracy is the labeled control arm

(57.1%), which is significantly higher than the unlabeled AI arm (45.8%). In the labeled-AI arm, the share of correct responses (51.6%) lies between them and is statistically indistinguishable from the two controls.

Overall, the manipulation performed as intended: the AI image elicited higher AI attributions once labeled. Yet respondents also frequently attributed AI to the authentic photograph, and labeling had only modest effects on correctness. Importantly, this is not due to inattention. The average dwell time on the article page exceeded 90 seconds in every arm with no systematic differences by treatment (Table B.2), indicating that the respondents spent time with the stimulus but still ex-post struggled to recall whether the image was synthetic or real. This aligns with work showing that photorealistic synthetic images are difficult to distinguish from real photojournalism (Nightingale and Farid, 2022). As the next section shows, these weak recognition responses matter: neither AI images nor labels reduce average engagement, but strong heterogeneity emerges depending on whether respondents believe the image is AI-generated or not.

3.2 Engagement: No average effect of AI or labels

Turning to behavior, Figure 4a plots newsletter demand by treatment arm, and Table B.4 reports the corresponding regressions. In the unlabeled control group, 36.5% of respondents request the newsletter. Assigning an AI image produces almost the same rate (37.0%), adding a label produces a small and insignificant increase (39.3%), and combining AI with a label yields 37.2%. In the regression, neither AI, nor Label indicators, nor their interaction is statistically different from zero. The joint test is also clearly insignificant.⁹

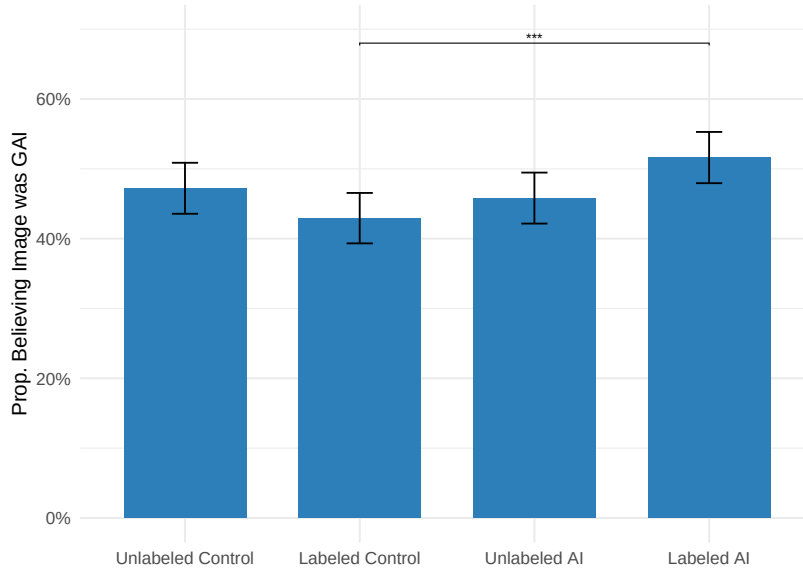
With $N = 2,870$ and $\alpha = 0.05$ (two-sided), the data exclude *moderate* average effects (larger than 5.5–7.8 p.p.). In similar experiments, Chopra et al. (2022) document very small average shifts (+1.4 p.p. for MSNBC among Democrats), which lie below our minimum detectable effect and are therefore consistent with our null. Chopra et al. (2024) likewise show treatment-induced changes of 5–9 p.p. within politically salient subgroups. Our precision thus rules out effects of the size commonly observed in subgroup analyses, while remaining compatible with near-zero changes in the aggregate sample.

Substantively, holding the article text and outlet constant, swapping a real photo for an AI image does not shift short-run demand. This speaks directly to current policy choices: newsroom labeling practices and platform-level disclosure standards (Nellis, 2024; Paul, 2024) are unlikely to affect behavior *unless* audiences actually notice and internalize the image source. This interpretation is reinforced by the weak recognition patterns documented above, and aligns with

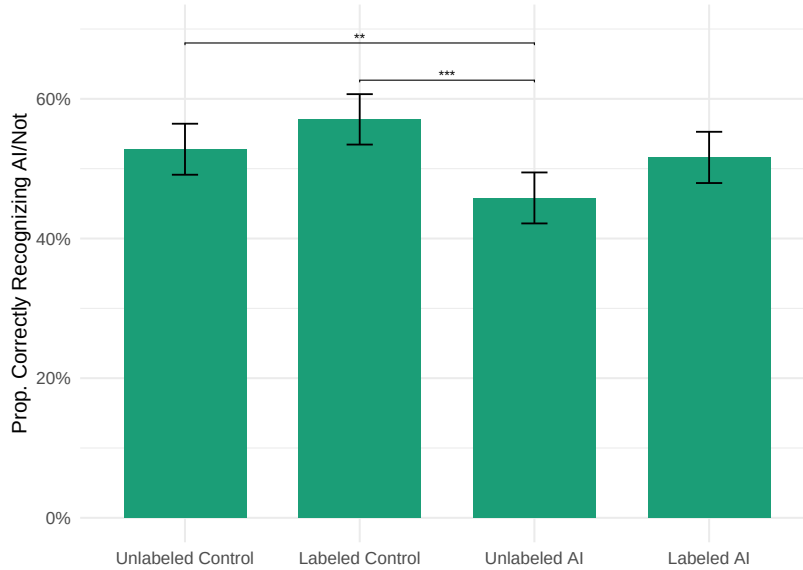
⁹Crucially, while inspecting the article, respondents were aware that they would later have the option to subscribe to the outlet’s weekly newsletter. They also spent over 90 seconds on the article page, with no systematic differences across arms (Table B.2), indicating balanced and sufficient attention. As a robustness check (not shown), we re-estimated all outcome effects controlling for reading time; results are unchanged. Likewise, no meaningful heterogeneity emerges above or below the median reading time.

Figure 3: Manipulation check results.

(a) Share Believing Image was AI Generated (*Manipulation = 1*) by Treatment Arm



(b) Share Identifying AI/Non-AI Correctly (*OriginCorrect = 1*) by Treatment Arm



Notes: Panel (a) shows the share of respondents reporting that the image was AI-generated (*Manipulation = 1*); Panel (b) shows the share who correctly recognized whether the image was AI-generated or not (*OriginCorrect = 1*). Error bars show 95% confidence intervals; stars above brackets denote significance levels (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$). Only significant pairwise comparisons are reported.

mixed evidence on what AI labels communicate and how users act on them (Altay and Gilardi, 2024; Wittenberg et al., 2025).

3.3 Heterogeneity: High demand with unnoticed AI, low if recognized

The average null demand effect masks pronounced heterogeneity depending on whether respondents correctly identified the image source (AI vs. not-AI). Figure 4b plots newsletter demand for respondents who correctly identified the image source ($OriginCorrect = 1$), while Figure 4c reports the same outcome for those who did not ($OriginCorrect = 0$). The corresponding regressions appear in Table B.4 (using triple interaction) and Table B.5 (using separate sub-samples).

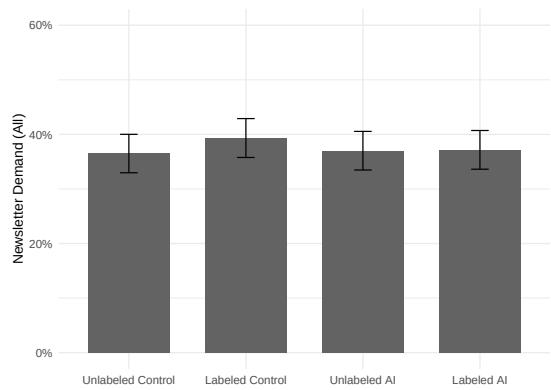
Among respondents who *did not* correctly identify the image source, newsletter demand in an AI image condition is by about 10 percentage points (0.081–0.105, $p < 0.01$) higher than with real image. Among those who *did* correctly identify the image source, newsletter demand instead is lower by a similar amount in the case of the AI image (–0.097 to –0.090, $p < 0.01$) as compared to the real image. These estimates come from the split-sample regressions in Table B.5, and the non-overlapping confidence intervals in both figures (4b and 4c) visually reinforce the sign reversal.

This pattern is formalized in a simple Bayesian updating framework (Appendix Section A). When respondents correctly identify the image source as AI, they revise their beliefs about outlet quality downward; when they do not, they revise beliefs upward. Demand then moves with these beliefs: higher when the image is not believed to be AI, and lower when it is. Because, in every treatment arm, roughly half of the respondents correctly identify the image and half do not, the positive and negative responses cancel out. As a result, the average effect of each treatment is close to zero, even though the underlying behavioural responses are large.

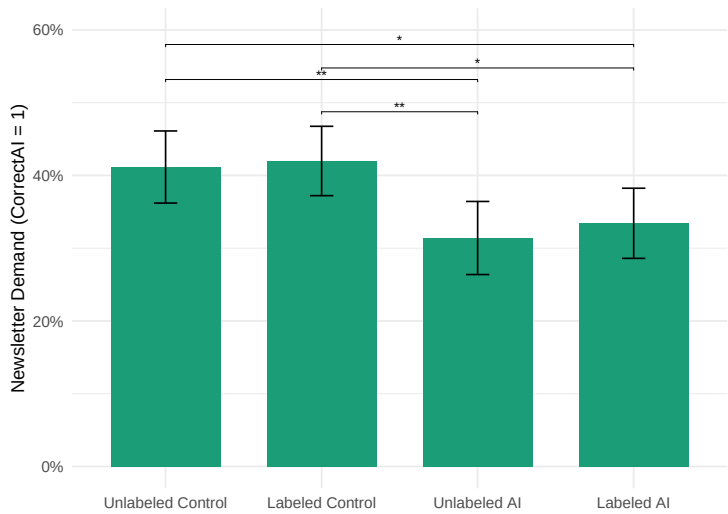
Labeling does not close the gap between respondents who correctly and incorrectly identified the image source. In the full-sample regression (Table B.4), the three-way $AI \times Label \times OriginCorrect$ interaction is small and statistically insignificant, and the $AI \times Label$ coefficients remain imprecise within each recognition subsample (Table B.5). Substantively, what matters is not whether the image is actually AI, but whether respondents believe it is: AI images sustain engagement when they go unnoticed and penalize the outlet when recognized, and labels do little to correct mistaken attributions. This pattern aligns with evidence that photorealistic synthetic images can appear trustworthy when indistinguishable from real photographs (Nightingale and Farid, 2022) and that AI visuals can shift perceptions in political news contexts (Caprini, 2024; Park et al., 2024).

Figure 4: Newsletter Demand by Treatment and AI Recognition Accuracy

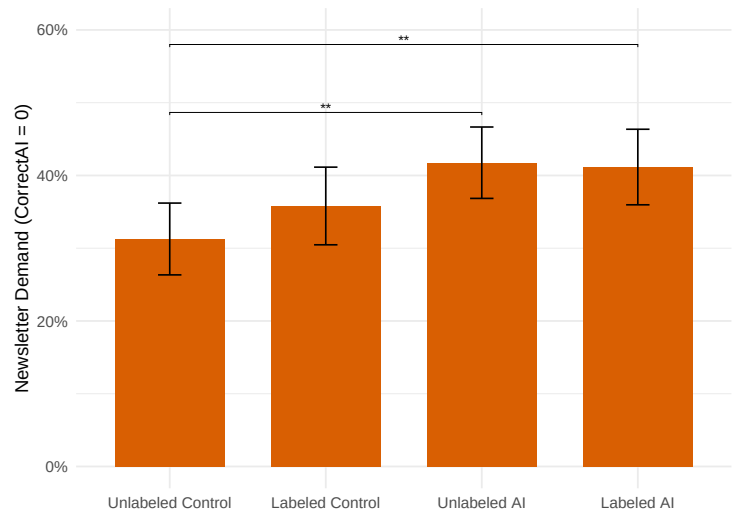
(a) All respondents



(b) Correct AI/Not (*OriginCorrect* = 1)



(c) Incorrect AI/Not (*OriginCorrect* = 0)



Notes: Each panel shows the share of respondents demanding the newsletter by treatment arm. Panel (a) includes all respondents; Panels (b) and (c) split the sample by whether respondents correctly identified whether the image was AI-generated (*OriginCorrect* = 1) or failed to do so (*OriginCorrect* = 0). Error bars represent 95% confidence intervals; stars above brackets denote significance levels (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$). Only significant pairwise comparisons are reported.

Another way to see that beliefs drive behavior, and that labels do not meaningfully change outcomes, is in Figures C.1a and C.1b. The figure stratifies newsletter demand solely by whether respondents believed the image was AI-generated (regardless of whether that belief was correct). Two patterns emerge. First, within each belief group — those who believe the image is AI and those who believe it is not — there is no systematic difference across treatment arms: assignment to AI versus control does not move demand conditional on belief. Second, the gap between belief groups is stark. Respondents who believed the image was AI exhibit substantially lower demand than those who believed it was real. This complements Figure 4: treatment assignment generates heterogeneity only insofar as it shifts beliefs, not because AI images or labels have an intrinsic direct effect. In other words, the behavioral differences are belief-driven: what people think they saw matters more than what they were shown. The flat treatment effects within belief groups reinforce that the heterogeneity arises from belief updates rather than from the experimental arm itself.

Two implications follow. *First*, the image source (AI vs. real) does not move average engagement unless users internalise it; recognition is crucial. *Second*, disclosure helps at the margin but is not a remedy, consistent with mixed evidence on labeling regimes (Altay and Gilardi, 2024; Wittenberg et al., 2025) and with platforms’ current rollouts (Nellis, 2024; Paul, 2024). For outlets, this creates a trade-off: AI visuals may help when they appear authentic, but may harm trust and engagement when recognised as synthetic, and labels only modestly reduce that risk.

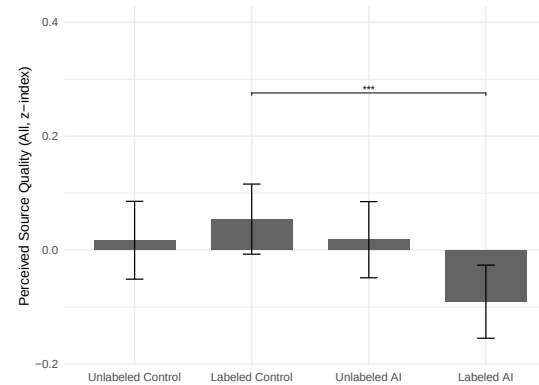
3.4 Mechanism: Credibility and engagement move together

Perceived news source quality mirrors the heterogeneous demand findings. Figure 5a shows that, pooling all respondents, average credibility barely differs across treatment arms: only the labeled AI image is rated significantly lower than the labeled control. This masks sharp differences once we separate respondents by whether they correctly identified the image source.

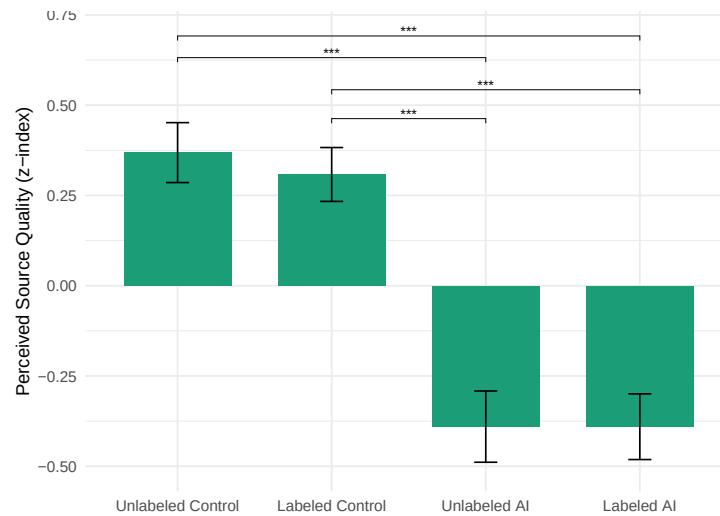
Table B.6 reports regressions for three sets of outcomes: the 1–5 score for *Accuracy*, *Trustworthiness*, and *Overall quality* (columns 7–12); their binary counterparts indicating whether respondents selected one of the two lowest categories (columns 1–6); and a composite credibility index defined as the average of the three standardized items (columns 13–14). Figure 5b and Figure 5c display the same composite graphically for respondents who correctly and incorrectly identified the image source, respectively.

Figure 5: Perceived Outlet Quality by Treatment and AI Recognition Accuracy

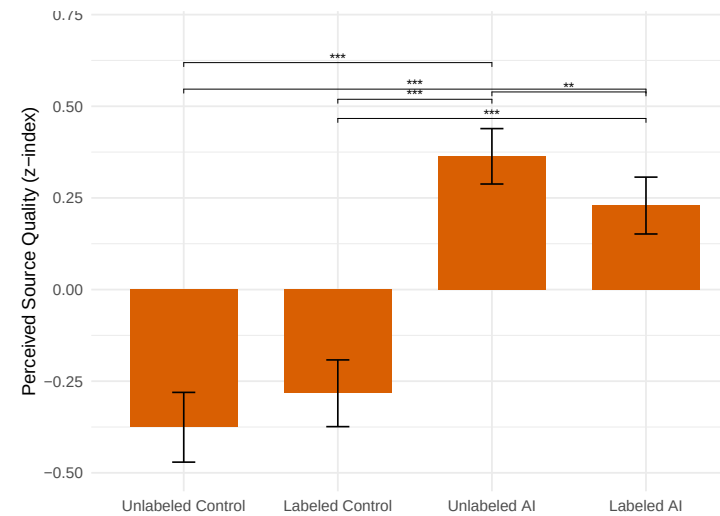
(a) All respondents



(b) Correct AI/Not (*OriginCorrect* = 1)



(c) Incorrect AI/Not (*OriginCorrect* = 0)



Notes: Each panel shows the average perceived outlet quality by treatment arm. Perceived outlet quality is a composite index averaging standardized responses to three 1–5 scale items: *Accuracy*, *Trustworthiness*, and *Quality*. Panel (a) includes all respondents; Panels (b) and (c) split the sample by whether respondents correctly identified whether the image was AI-generated (*OriginCorrect* = 1) or failed to do so (*OriginCorrect* = 0). Error bars represent 95% confidence intervals; stars above brackets denote significance levels (***) $p < 0.01$, (**) $p < 0.05$, (*) $p < 0.10$). Only significant pairwise comparisons are reported.

Among respondents who *did not* correctly identify the image source (*OriginCorrect* = 0), the AI image raises perceived credibility (consistent with believing it was a real photograph; see Figure 5c): the composite index increases by roughly 0.6–0.7 standard deviations relative to the control arm (columns 13–14, $p < 0.001$), and each 1–5 item in columns 7–12 shifts upward by a similar margin. Among respondents who *did* recognise the image as AI (*OriginCorrect* = 1), the pattern reverses (see Figure 5b). The interactions ($AI \times OriginCorrect$) in columns 13–14 are strongly negative (around -1.3 to -1.5 SD, all $p < 0.001$), implying net credibility losses of roughly -0.7 to -0.8 SD. Labeling only slightly softens these penalties: for some outcomes, the $AI \times Label$ coefficient is weakly negative; for others the triple interaction is weakly positive; but no specification brings perceived quality back to the non-AI baseline.

Viewed through a belief-updating lens, Figures C.1c and C.1d shows perceived outlet quality separately for respondents who believed the image was AI-generated versus those who believed it was not, irrespective of whether that belief was accurate. As with newsletter demand, treatment assignment does not meaningfully move perceived quality within either belief group: respondents who believe the image is real rate the outlet similarly across all treatment arms, and the same holds for those who believe it is AI. The large gap arises between belief groups, not within them. Believing the image is synthetic sharply lowers perceived accuracy, trustworthiness, and overall quality relative to believing it is real, even when both groups were exposed to the same stimulus.

Taken together, Figures 4, 5, and C.1 indicate that participants reward the outlet when they believe the image is authentic and penalise it when they believe it is synthetic — even when they are wrong. This credibility pattern mirrors the recognition-contingent demand effects reported in Section 3.3, supporting the interpretation that AI images influence engagement primarily by shifting beliefs about the news source, rather than through their intrinsic visual properties or the presence of a disclosure.

The joint movement of credibility and sign-ups is consistent with models in which perceived outlet quality drives news demand, and with experimental work using newsletter interest as a behavioral proxy (Chopra et al., 2024). In our setting, the operative mechanism is belief updating about the *news source*, conditional on whether respondents think the image is authentic, whether or not the image origin is fully disclosed.

As a complementary test, we treat reported AI recognition as endogenous and instrument it using the randomized $AI \times Label$ interaction (Table B.7). The first stage confirms that the interaction raises the likelihood of reporting the image as AI-generated by about 0.10 ($p < 0.01$), while the main effects of AI and Label alone are negligible. In the second stage, the 2SLS estimates for newsletter demand are negative but imprecise, consistent with the small reduced-form effects of $AI \times Label$. By contrast, the 2SLS estimates for perceived outlet quality are negative and statistically significant ($p < 0.05$), mirroring the reduced-form patterns: respondents pushed into believing the image was AI rate the outlet as less credible. Taken together, both reduced-form and IV evidence point to the same interpretation: AI images influence engagement

primarily through beliefs about source authenticity.¹⁰

We also test several alternative explanations (Table B.8). AI images do not increase perceived political bias or confusion about the text, and respondents do not disengage because they distrust the researcher. Average trust in the researcher remains above "somewhat trustworthy" in every arm, and while trust moves slightly across conditions, the only systematic declines appear when respondents recognise the image as AI (similar to what is observed for source credibility).¹¹ AI images are also rated as slightly more entertaining on average, with penalties emerging only when respondents recognise the image as synthetic. These patterns rule out concerns that the null average effect or the heterogeneous responses reflect scepticism about the setup or perceived manipulation.

Two additional pieces of evidence support this interpretation. First, open-ended answers about the study's purpose (Table B.9) suggest respondents did not guess the hypothesis we test, reducing concern about experimenter demand. A majority state that the study was about "AI or artificial intelligence" (53.5%) or "fake news/media" (24.5%). Very few mention trust or credibility (2.3%), and almost nobody suggests deception (0.1%). Their examples, e.g.: "*Whether humans can tell whether an article is generated by AI*" or "*Fake news*", indicate respondents took the task at face value and did not anticipate a credibility-focused hypothesis.

Second, open-ended reasons for requesting (or not requesting) the newsletter (Table B.10) align with a neutral, topic-driven interpretation rather than any suspicion. The modal category is curiosity or novelty (47%), followed by interest in AI/technology (33.5%). Very few mention trust or credibility (1.8%). Respondents write, for example, "*I enjoyed reading the article and it gave me a better insight into...*". These answers reinforce that participants took the task at face value and were responding to the content, not attempting to second-guess the experiment. A small minority even inferred that if the image was synthetic, the article text might also be AI-generated, for example: "*I believe the article is AI generated due to the sentence 'AI-generated image (created using OpenAI's DALL-E)' under the photo. I assume the whole article, not just the photo, is AI generated.*" This reinforces that respondents were updating beliefs about the news outlet and its content, not about the researcher or the study design, and only when they recognised the image as AI.

Taken together, alternative mechanisms, guessed study purpose, and open-ended motivations all point in the same direction: respondents interpreted the task straightforwardly, and the demand

¹⁰The Wu–Hausman tests do not reject exogeneity of reported AI recognition for either outcome, implying that the IV and OLS estimates are statistically indistinguishable; the IV results therefore reinforce the reduced-form interpretation.

¹¹This pattern does not support a deception-based explanation. If respondents distrusted the *experiment*, trust should fall for participants generally exposed to AI, especially in the unlabeled-AI arm, where concerns about being misled would be strongest. Instead, trust rises among respondents who do *not* detect the AI image and falls only among those who do, mirroring the credibility penalty applied to the news outlet. The most consistent interpretation is inference spillover: recognising an AI image lowers beliefs about the outlet and, by association, trust in the presenter of the article, rather than signalling broad experiment distrust.

and credibility effects are driven by whether they believe the image is authentic, not by confusion, scepticism, or experimenter demand.

4 Experiment 2: Robustness and Mechanisms

A central conclusion of Experiment 1 is that outcomes respond primarily to beliefs about authenticity: respondents who believe the image is AI exhibit substantially lower demand and lower perceived outlet quality, while demand can prove higher when AI imagery goes unnoticed, producing an aggregate null via offsetting belief updates.

Two potential concerns can be raised about Experiment 1: (i) that the results may be specific to the single image and (ii) that the belief-based heterogeneity whether the image is AI we document is not causal. To rule out these concerns, we fielded a second survey experiment that preserves the core structure of our original design while adding new article–image stimuli and introducing randomized variation in *recognizability* among AI images. Comparing *AI very realistic* to *AI not very realistic* images within the AI origin arms shifts the probability that respondents believe the image as AI while holding constant (i) the semantic scene and (ii) the fact that the image is AI-generated.

4.1 Article stimuli construction

Each respondent views exactly one article sampled from a fixed set of 6 news articles. The articles cover diverse topics from the same news outlet (Business + Economy, Politics + Society, Arts + Culture, Education, Environment). Articles are again selected from the same online outlet: newspaper *The Conversation*.

For each respondent, one article from this set is drawn with equal probability. Conditional on the selected article, respondents are randomly assigned to one of six image conditions described below. Because our research question concerns responses to image authenticity rather than heterogeneity across article content, we treat the article dimension as a nuisance parameter. In the main analyses, we pool observations across articles and estimate average treatment effects of image origin, labeling, and recognizability, absorbing any level differences across articles via article fixed effects.

4.2 Image construction and AI generation protocol

Each article is paired with one original real photograph that accompanied the article in its source context. From this photograph, we construct two AI-generated versions intended to depict the same underlying scene and composition. The goal is to hold semantic content and layout constant

while varying perceived authenticity.

AI images are generated using a two-step protocol designed to separate description from generation. First, on a dedicated instance, we provide ChatGPT 5.2 with the article page and instruct it to produce a detailed, literal description of the accompanying photograph, including objects, setting, composition, perspective, lighting, and salient visual elements, while avoiding any mention of provenance, realism, or artistic style. Second, on a separate instance of ChatGPT Image (with no access to the original photograph), we provide the textual description from the previous step and instruct the model to generate an image matching the described scene and composition.

At this generation stage, we manipulate photorealism to vary the recognizability of AI imagery. One version is prompted to be highly photorealistic (*AI very realistic*), closely resembling a professional news photograph. The other is prompted to be intentionally less photorealistic (*AI not very realistic*), with visual cues that make AI generation more apparent. Across all variants, the depicted scene, composition, and semantic content are held constant.

4.3 Treatment arms

The experiment follows a between-subjects design with randomized variation in image origin, labeling, and recognizability. Respondents are assigned to one of six conditions:

1. **Control, no label:** real photograph with no credit or disclosure line.
2. **Control, label:** real photograph with a conventional wire-service credit line.
3. **Hard AI, no label:** highly photorealistic AI-generated image with no disclosure line.
4. **Hard AI, label:** same highly photorealistic AI-generated image with an “AI-generated image” disclosure line.
5. **Easy AI, no label:** non-photorealistic AI-generated image with no disclosure line.
6. **Easy AI, label:** same non-photorealistic AI-generated image with an “AI-generated image” disclosure line.

The first four conditions replicate the origin $\times$ label design of Experiment 1 on a different image set, addressing concerns about reliance on a single picture. The additional two conditions introduce exogenous variation in the recognizability of AI imagery while maintaining the AI origin of the images fixed. This extension allows us to isolate the causal role of perceived authenticity and belief formation, the mechanism emphasized in our main results.

4.4 Survey procedure and outcome measures

After providing informed consent, respondents complete standard attention and quality checks, followed by baseline demographics and media-use covariates. Respondents then view the article page, including the assigned image, for a minimum enforced dwell time¹².

Immediately after exposure, respondents complete the primary behavioral outcome *before* any recognition or disclosure-related questions. Specifically, they indicate whether they would like to receive the outlet’s newsletter. We code newsletter demand as an indicator equal to one if the respondent requests the newsletter and zero otherwise. Eliciting this decision prior to recognition questions avoids mechanically increasing scrutiny of image authenticity and reduces the scope for post-treatment measurement to confound respondents’ behavioral response.

After the newsletter decision, respondents report whether they believe the image was AI-generated. We then measure perceived outlet quality using the same items as in Experiment 1—accuracy, trustworthiness, and overall quality—combined into a standardized index.

We also elicit broader beliefs about AI use in journalism, measured with two survey questions on the prevalence of AI-generated text and images. To minimize priming, questions on **AI familiarity**—including prior usage, knowledge, and exposure—are placed at the end of the survey.

4.5 Sample and implementation

The study was conducted online using Prolific and pre-registered on OSF with the number [e5ncv](#). Respondents completed a CAPTCHA, an attention check, and a screening question related to the use of large language models. Participants failing any pre-registered quality checks were excluded. We have also enabled LLM and BOT authenticity checks using the integrations between Qualtrics and Prolific.

After exclusions, the final sample consists of 4,196 respondents. Prolific’s built-in speed check was enabled to exclude respondents completing the survey unreasonably fast relative to the median, ensuring sufficient time-on-task. The median completion time in the final sample was eight minutes. Respondents were paid £2 (over £14 per hour), consistent with Prolific guidelines.

Table [B.11](#) shows that observable characteristics are balanced across experimental conditions. The sample is broadly similar, though somewhat more educated and with higher average income.

¹²30 seconds.

5 Results from Experiment 2

We now turn to results from Experiment 2, which extends the core design of Experiment 1 in two ways: it replicates the origin $\times$ label manipulation across six diverse news articles, and introduces exogenous variation in AI photorealism to provide a causal test of the recognition channel. We first confirm that the manipulation of image origin and photorealism successfully shifts AI recognition across arms, before showing that the aggregate null on newsletter demand replicates across stimuli and that the photorealism instrument isolates the causal effect of recognition on outlet quality.

5.1 Replication over Multiple Articles

Recognition. Figure 6a, Figure 6b, and Table B.12 show that our manipulation of image origin and photorealism level successfully shift correct AI recognition across arms. In the control conditions, approximately 63.3% of respondents correctly identified the original photograph as not AI-generated; the wire-service label added virtually nothing (63.9% in the labeled control; insignificant coefficient of 0.006). Among respondents assigned the less photorealistic AI image, correct origin recognition rises sharply — to 81.2% in the unlabeled easy-AI arm and 86.5% in the labeled easy-AI arm, gains of roughly 18 and 23 percentage points over the unlabeled control (both $p < 0.001$). The pattern reverses for the highly photorealistic AI image: only 44.1% of respondents in the unlabeled hard-AI arm correctly identified it as AI-generated, well below the control baseline, rising modestly to 52.0% with a label (a significant improvement of 7.3 p.p., $p < 0.05$, but still far below the easy-AI arms).

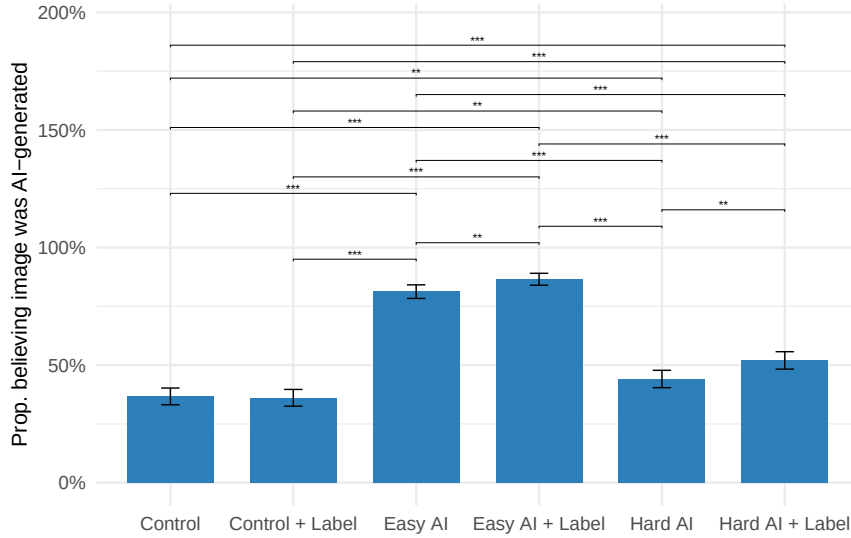
These results confirm that photorealism is the primary driver of recognition, while labels offer only modest correction regardless of image type. The gap between easy-AI and hard-AI arms is large and precisely estimated — roughly 37 percentage points in the unlabeled conditions. The hard-AI conditions closely mirror Experiment 1, where correct recognition in the unlabeled and labeled AI arms stood at 46% and 52%, respectively. The control arms are somewhat more accurate in Experiment 2 (63% versus 53%), plausibly due to random variation in recognition across the richer set of six-article stimulus, making authentic photographs easier to identify on average. The close alignment between hard-AI arms across both experiments — despite different articles, images, and recruitment waves — confirms that the recognition patterns are not idiosyncratic to a single image.

Tables B.12 (OLS) and B.13 (Logit) report treatment effects on two binary outcomes: *Manipulation* (equal to 1 if the respondent identified the image as AI-generated) and *OriginCorrect* (equal to 1 if the respondent correctly identified the image’s origin). All specifications include image fixed effects; even-numbered columns add demographic controls.

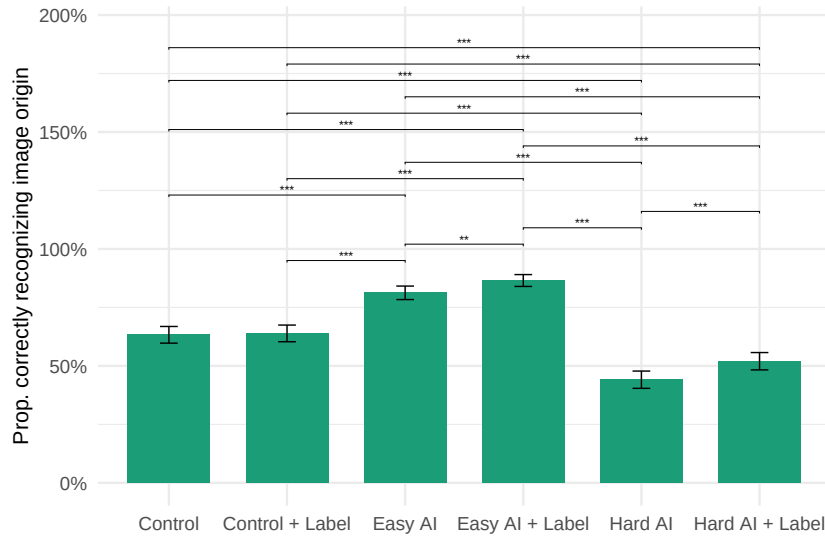
The easy-AI arms drive the largest shifts in both outcomes: relative to the control, assignment

Figure 6: Manipulation check results.

(a) Share Believing Image was AI Generated (*Manipulation = 1*) by Treatment Arm



(b) Share Identifying AI/Non-AI Correctly (*OriginCorrect = 1*) by Treatment Arm



Notes: Panel (a) shows the share of respondents reporting that the image was AI-generated (*Manipulation = 1*); Panel (b) shows the share who correctly recognized whether the image was AI-generated or not (*OriginCorrect = 1*). Error bars show 95% confidence intervals; stars above brackets denote significance levels (***) $p < 0.01$, ** $p < 0.05$, * $p < 0.10$). Only significant pairwise comparisons are reported.

to unlabeled easy-AI raises *Manipulation* by 44.6 percentage points and *OriginCorrect* by 17.7 percentage points (both $p < 0.001$), with the easy-AI + label arm adding a further 5.1 points to *Manipulation* ($p < 0.05$) and 4.8 points to *OriginCorrect*. The hard-AI arms tell a complementary story: assignment to unlabeled hard-AI raises *Manipulation = 1* by just 7.3 percentage points

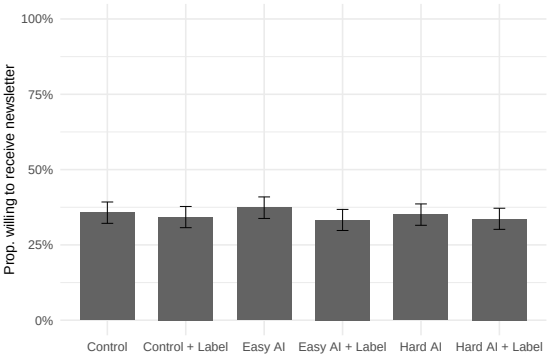
($p < 0.001$), consistent with the image’s visual similarity to a real photograph, implying a drop in *OriginCorrect* = 1 by 19.4 percentage points ($p < 0.001$). Adding a label partially corrects this (the hard-AI $\times$ label interaction on *OriginCorrect* is $+7.6p.p.$, $p < 0.05$), however correct recognition remains significantly below the easy-AI conditions. The labeled control, by contrast, produces no significant change in either outcome relative to the unlabeled control, confirming that a wire-service credit does not meaningfully shift attributions for authentic photographs. Together, these tables validate our photorealism manipulations as a strong and credible source of exogenous variation in recognition.

Demand for Newsletter Figure 7a and Table B.14 replicate the aggregate null result from Experiment 1. Baseline newsletter demand in the unlabeled control is 35.7%, and estimated differences across all other arms are small and statistically indistinguishable from zero: -1.5 p.p. in the labeled control, $+1.7$ p.p. in the unlabeled easy-AI arm, -2.4 p.p. in the labeled easy-AI arm, -0.6 p.p. in the unlabeled hard-AI arm, and -2.0 p.p. in the labeled hard-AI arm (none significant; joint test also clearly insignificant). These results hold with and without article fixed effects (columns 7–9), confirming this null result is not driven by compositional differences across the six articles. Behind the aggregate null, Table B.14 reveals the same sharp heterogeneity documented in Experiment 1, now precisely and causally identified thanks to the photorealism manipulation (i.e., using a more or less recognizable AI pictures, while reproducing the same content as in the original wire-photo). Column (9-12) interact each treatment indicator with one to determine whether the respondent correctly identified the origin of the image (*OriginCorrect*).

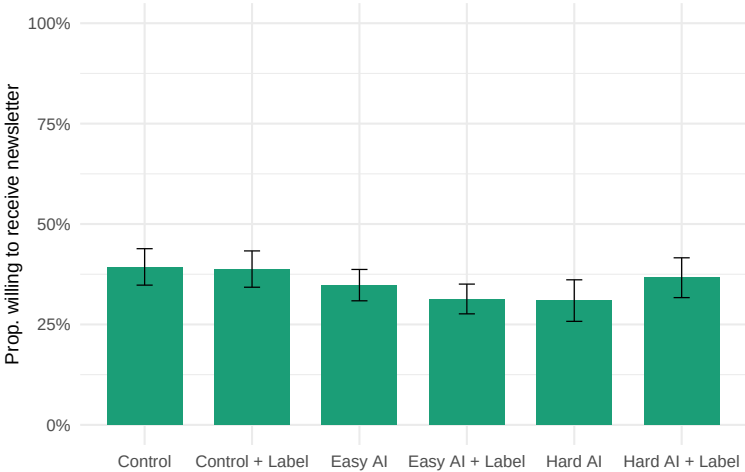
Among respondents who *did not* correctly recognize the image origin — that is, those shown an AI-generated image who believed it was real (or vice versa) — newsletter demand rises substantially relative to the control conditions: $+19.5$ p.p. in the unlabeled easy-AI arm ($p < 0.001$), $+16.4$ p.p. in the labeled easy-AI arm ($p < 0.001$), and $+8.9$ p.p. in the unlabeled hard-AI arm ($p < 0.05$). The labeled hard-AI arm is the only exception, with a near-zero coefficient of $+0.9$ p.p., consistent with seeing this as the original article with no AI generated imagery. Among respondents who *did* correctly identify the image as AI-generated, the pattern reverses sharply. The interactions (treatment $\times$ *OriginCorrect*) are negative and significant, depressing demand relative to the control for the unlabeled easy-AI arm (-0.240 , $p < 0.001$), for the labeled easy-AI arm (-0.244 , $p < 0.001$), and for the unlabeled hard-AI arm (-0.174 , $p < 0.001$). Adding both main effect and interaction to the *OriginCorrect* baseline of $+0.098$ yields net demand changes of approximately -4.5 p.p., -8.0 p.p., and -8.5 p.p. relative to the control for these three arms, respectively. The labeled hard-AI arm again stands apart: its interaction is small and insignificant (-0.035), implying little or no demand penalty among correctly recognizing respondents — consistent with the label partially legitimizing AI use when the image is difficult to detect as synthetic.

Figure 7: Newsletter Demand by Treatment and AI Recognition Accuracy

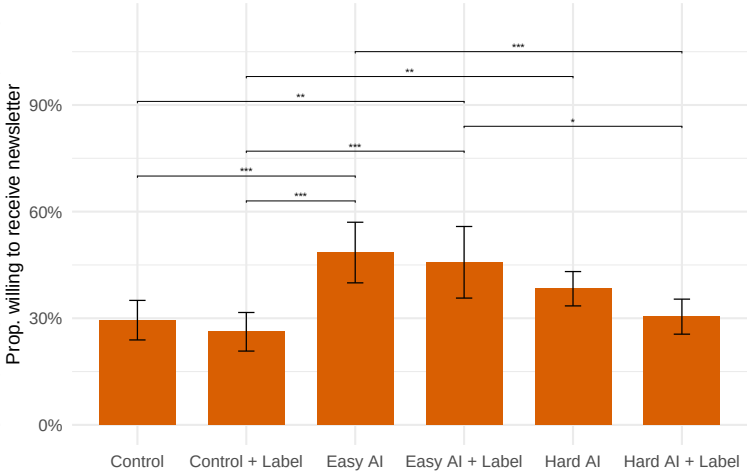
(a) All respondents



(b) Correct AI/Not (*OriginCorrect* = 1)



(c) Incorrect AI/Not (*OriginCorrect* = 0)



Notes: Each panel shows the share of respondents demanding the newsletter by treatment arm. Panel (a) includes all respondents; Panels (b) split the sample by whether respondents correctly identified whether the image was AI-generated (*OriginCorrect* = 1) or failed to do so (*OriginCorrect* = 0). Error bars represent 95% confidence intervals; stars above brackets denote significance levels (***) $p < 0.01$, (**) $p < 0.05$, (*) $p < 0.10$). Only significant pairwise comparisons are reported.

Columns (5-8) stratify instead by respondents’ *belief* about whether the image was AI-generated (*Manipulation*), regardless of accuracy. Believing the image is AI reduces demand by 9.8 p.p. ($p < 0.001$) uniformly across treatment arms, confirming that subjective perception of AI use — not actual image origin — drives behaviour. The treatment $\times$ *Manipulation* interactions are small and insignificant for most arms. The notable exception is the labeled hard-AI arm, where the interaction is +0.161 ($p < 0.001$), substantially offsetting the baseline “believing is AI” penalty. In this case, the net demand effect is approximately $16.1 - 9.8 = 6.3$ p.p. This suggests that a label on a hard-to-detect AI image can reframe audience inference, communicating transparency rather than carelessness, but only when the image itself provides few visual cues that would independently trigger suspicion.

Both sets of columns point to the same conclusion: the aggregate null arises from precisely offsetting belief-driven responses, and what governs engagement is not whether an image is actually AI-generated, but whether audiences come to believe it is.

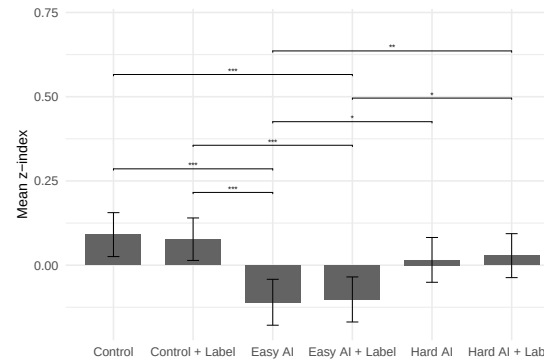
Perceived Quality Figure 8a and Table B.15 show that perceived outlet quality closely tracks the recognition-contingent pattern of newsletter demand. In the pooled sample, AI conditions appear to either leave unchanged or marginally worsen the perceived quality of the outlet relative to the control. This average, however, masks sharp heterogeneity by recognition. Among respondents who *did not* correctly identify the image origin — that is, respondents shown an AI-generated image who believed it was a wire-service photograph — all AI conditions exhibit higher perceived quality by roughly 0.5 to 0.6 standard deviations than the control article-pictures (Figure 8b), consistent with Experiment 1. Among respondents who *did* correctly identify the image as AI-generated, the pattern reverses: the AI $\times$ *OriginCorrect* interactions in Table B.15 are all large and negative (ranging from -0.96 to -1.29 SD, all $p < 0.001$), implying net credibility losses of approximately -0.3 to -0.7 SD relative to the control.

This heterogeneous pattern holds uniformly across both easy-AI and hard-AI conditions: while photorealism governs the likelihood of recognition, once AI images are recognized, the credibility penalty is comparably large regardless of whether the image was easy or hard to detect as synthetic. Labels offer no meaningful attenuation, with label interactions small and statistically indistinguishable from zero across all quality outcomes. Among correctly recognizing respondents, the hard-AI arm attracts a marginally larger quality penalty (-1.26 SD) than the easy-AI arm (-1.17 SD), though the difference is not precisely estimated.

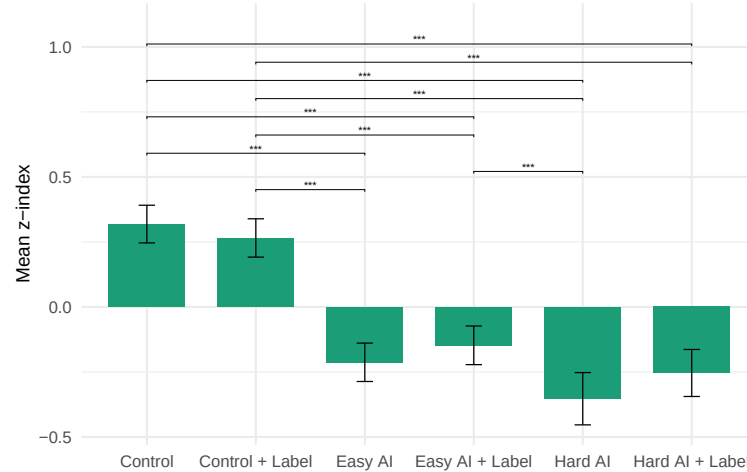
Table B.15 reports OLS estimates of treatment effects on the composite *z-index* (Column 1), which averages standardized responses to perceived *Accuracy*, *Trustworthiness*, and *Quality* of the outlet, with higher values indicating worse perceptions. The dichotomous versions of these variables are presented in Columns 2–4, alongside secondary outcomes including *Biased*, *VeryComplex*, and *NotEntertaining* in Columns 5–8.

Figure 8: Perceived Outlet Quality by Treatment and AI Recognition Accuracy

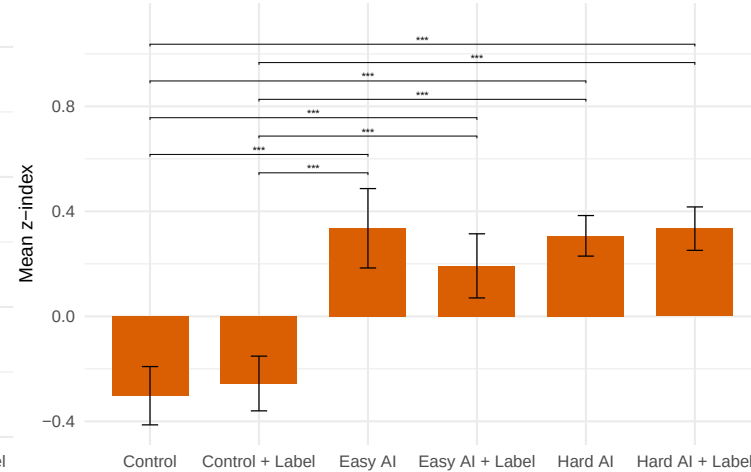
(a) All respondents



(b) Correct AI/Not (*OriginCorrect* = 1)



(c) Incorrect AI/Not (*OriginCorrect* = 0)



Notes: Each panel shows the average perceived outlet quality by treatment arm. Perceived outlet quality is a composite index averaging standardized responses to three 1–5 scale items: *Accuracy*, *Trustworthiness*, and *Quality*. Panel (a) includes all respondents; Panels (b) split the sample by whether respondents correctly identified whether the image was AI-generated (*OriginCorrect* = 1) or failed to do so (*OriginCorrect* = 0). Error bars represent 95% confidence intervals; stars above brackets denote significance levels (***) $p < 0.01$, (**) $p < 0.05$, (*) $p < 0.10$). Only significant pairwise comparisons are reported.

At baseline, among individuals who do not correctly recognize the image origin (*OriginCorrect* = 0), all four AI treatment arms significantly improve perceived outlet quality relative to the control: *Easy AI* by 0.68 SD, *Easy AI + Label* by 0.50 SD, *Hard AI* by 0.62 SD, and *Hard AI + Label* by 0.66 SD (all $p < 0.001$). Among respondents who correctly identify the image as AI-generated, the opposite is true: the *OriginCorrect* interactions are -1.20 SD for *Easy AI*, -0.96 SD for *Easy AI + Label*, -1.29 SD for *Hard AI*, and -1.22 SD for *Hard AI + Label* (all $p < 0.001$), with implied net credibility losses relative to the control harm of approximately -0.3 to -0.7 SD across conditions.

By contrast, *Biased* and *VeryComplex* are largely unaffected by treatment or recognition, ruling out a general scepticism spillover onto article content. The one positive secondary effect is on *NotEntertaining*, where AI conditions reduce this indicator, suggesting that recognised AI imagery may attract novelty interest without translating into higher engagement. These patterns closely replicate the mechanism results from Experiment 1, extending them to a multi-article, multi-image design and confirming that the credibility penalty is belief-driven rather than contingent on any specific visual stimulus.

5.2 Causal Mechanism: The Role of Recognition

In Experiment 2 we introduce exogenous variation in AI photorealism to get a clean IV for recognition. We here test the role of recognition as a causal channel for the change in news demand and perceptions of outlet standing.

The identification strategy exploits the fact that easy-to-recognise and hard-to-recognise AI images shift the probability of correct recognition while holding image origin and depicted scene constant. We proceed in three steps. First, we verify that the photorealism manipulation is a strong first stage. Second, we show that instrumenting for recognition yields a precise null on newsletter demand, consistent with the aggregate null arising from symmetric, offsetting belief updates rather than the absence of any belief-driven mechanism. Third, and more sharply, we find that recognizing AI causally reduces perceived outlet quality by 0.35 standard deviations — an effect that is targeted at perceptions of credibility and does not spill over into perceptions of article bias or complexity.

Varying AI recognizability The heterogeneity results in Section 5.1 establish that newsletter demand and perceived outlet quality move sharply depending on whether respondents correctly identify the image origin. However, the recognition indicator (*OriginCorrect*) is elicited after treatment exposure and may therefore reflect post-treatment selection rather than a causal belief mechanism. Experiment 2 addresses this concern directly: randomly assigning respondents to easy-to-recognise versus hard-to-recognise AI images shifts the probability of correct recognition while holding image origin and depicted scene constant. We use this exogenous variation in

recognizability due to random treatment assignment as an instrument for *OriginCorrect* and estimate the causal effect of correctly identifying the image source on newsletter demand and perceived outlet quality.

First stage The photorealism manipulation is a strong instrument. As reported in Table B.16, the robust partial F -statistic for the excluded instrument reaches 171.1 when the endogenous variable is *Manipulation* and 91.2 when it is *OriginCorrect*, both far exceeding conventional weak-instrument thresholds. Respondents assigned the less photorealistic AI image are substantially more likely to identify the image as AI-generated (see also Figure 6 and Table B.12), generating the first-stage variation needed to isolate the causal role of recognition.

Newsletter demand The 2SLS estimates for newsletter demand, reported in Table B.16 (columns 1–4, are small and statistically indistinguishable from zero: the instrumented coefficient on *Manipulation* is +0.007 (s.e. 0.036) and on *OriginCorrect* is +0.015 (s.e. 0.050). The confidence intervals around these estimates — which do not rule out effects of up to 8–11 percentage points in either direction — are consistent with the aggregate null on demand documented in Section 5.1, and with the interpretation that positive and negative belief-driven responses to recognition offset each other in the aggregate, leaving the average effect close to zero.

Perceived outlet quality The causal evidence is sharper for perceived outlet quality. Table B.16 (columns 5–8) reports 2SLS estimates for the composite quality index: the instrumented coefficient on *Manipulation* is -0.366 (s.e. 0.066, $p < 0.001$) and on *OriginCorrect* is -0.348 (s.e. 0.091, $p < 0.001$), both robust to the inclusion of controls. These estimates provide clean causal evidence that recognition — not mere exposure to an AI image — is the operative mechanism depressing credibility: correctly identifying the image as AI-generated reduces perceived outlet quality by approximately 0.35–0.37 standard deviations.

Table B.17 further instruments *OriginCorrect* with the photorealism manipulation and breaks the quality index down into its constituent dimensions. The effect is visible across all three quality dimensions: perceived accuracy falls (coefficient on *NotAccurate*: +0.108, $p < 0.001$), trustworthiness declines (*NotTrustworthy*: +0.084, $p < 0.05$), and overall quality deteriorates (*LowQuality*: +0.126, $p < 0.001$). By contrast, perceived political bias and article complexity are unaffected (coefficients of -0.046 and $+0.001$, both insignificant), ruling out the possibility that recognition triggers a more general scepticism about the article’s content rather than a targeted inference about the outlet’s editorial standards. The one positive effect is on entertainment value: recognition reduces the probability that respondents rate the article as not entertaining (-0.166 , $p < 0.001$), consistent with correctly identified AI imagery attracting curiosity or novelty value that partially offsets the credibility loss, but without translating into

higher engagement.

These IV results complement and extend the findings from Experiment 1, where the instrument was the randomised AI $\times$ Label interaction and the 2SLS estimate for quality was negative and significant while the demand estimate was imprecise (Table B.7). Here, the stronger and cleaner photorealism instrument delivers precisely estimated negative effects on perceived outlet quality across both instrumented variables, supporting the interpretation that recognition causally depresses outlet credibility and that the null effect on demand arises from symmetric, offsetting belief updates — higher among those who do not recognise the image as AI, lower among those who do — rather than from the absence of any belief-driven mechanism.

6 Journalists' Beliefs

To assess external validity, we conducted a survey with 132 international journalists recruited via Prolific. The sample had an average age of 35.1 years and skewed female (59.8%). Respondents were spread across multiple countries, with the United States and the United Kingdom being the two most represented markets (35.1% and 21.1% respectively). In terms of outlet type, online newspapers were the most common workplace (29.2%), followed by freelance journalists (26.2%). The vast majority were currently working as journalists (76.5%), most employed full time (71.2%), and the sample skewed heavily toward experienced professionals, with over 80% having worked in the sector for more than five years. Editors and reporters together accounted for over 80% of the sample. Full sample characteristics are reported in Table B.18.

In this survey, we elicited journalists' beliefs about how using AI-generated images in online articles affects demand for news for that outlet. We did so by leveraging results from Experiment 2. First, we asked respondents about the newsletter sign-up rate in the Control group and then showed it to them as an anchor to calibrate beliefs about baseline demand. We then elicited journalists' beliefs about the newsletter sign-up rate when the article included either a Hard-to-Recognize AI image or an Easy-to-Recognize AI image (presented in random order to mitigate order effects). Next, we elicited beliefs about sign-up rates under the same image conditions when a label was also shown. Finally, we incentivized accuracy using data from Experiment 2: we informed journalists that they would receive an additional \$0.50 bonus if their answers exactly matched the Experiment 2 results.

Figure 9a reveals that journalists' beliefs about the effect of AI-generated images on newsletter sign-up rates exhibit a systematic miscalibration and a striking disconnect with the actual experimental results. Journalists believed that approximately 52 out of 100 readers would sign up in the control condition, substantially overestimating the actual sign-up rate of 35 observed in Experiment 2. Beyond this baseline miscalibration, journalists expected AI-generated images to meaningfully depress demand. Anticipated sign-up rates fell to 38.5 for hard-to-recognize AI

images and further to 24.4 for easy-to-recognize ones — a gap of nearly 14 percentage points suggesting that journalists believe image recognizability plays an important role in shaping reader behavior. They also viewed labeling as consequential: in the hard-to-recognize condition, the expected sign-up rate fell further to 28.8 when a label was added, while beliefs in the easy-to-recognize condition remained largely unchanged at 23.2. Crucially, however, none of these anticipated effects are borne out by the data: Experiment 2 finds no statistically significant difference in demand across any of the image conditions. Journalists thus not only overestimate baseline engagement, but also overestimate the extent to which readers notice, and respond to, the AI origin of images.

We further explored journalists' views on using AI to generate images and write articles, as well as their perceptions of how widespread this behaviour is among their colleagues. Figure 9b shows that journalists display strong personal support for transparency around AI-generated content: over 90% agreed that AI use should be clearly disclosed in articles, with only a small minority expressing any disagreement. Views on personal AI adoption are more divided, with roughly equal shares reporting having used or never used AI in their work.

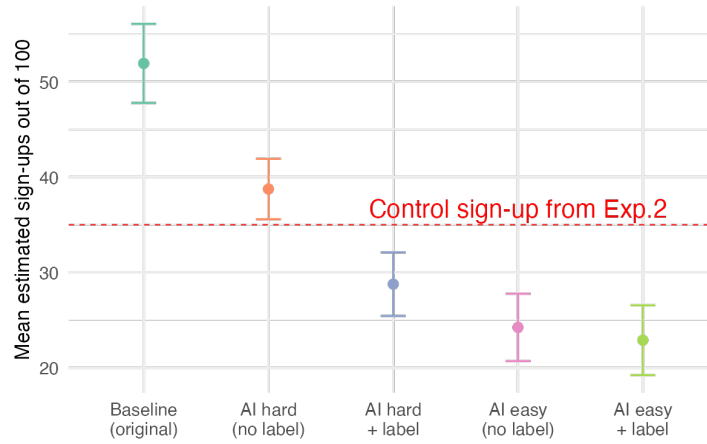
Figure 9c shows journalists' second-order beliefs about their peers, which reveal a degree of social projection. The distribution of estimated peer usage of AI is wide and dispersed across the full 0–100 range, with a mean of approximately 50 out of 100 — somewhat higher than journalists' own self-reported behaviour would suggest. By contrast, estimated peer support for disclosure is more concentrated toward higher values, yet centres around 64 out of 100 on average, well below the over 90% observed in the sample itself, suggesting that journalists substantially underestimate how widely pro-disclosure norms are shared among their peers.

When asked why they believed journalists use AI in their work, respondents identified a range of motivations, with efficiency and research support emerging as the two most prominent.¹³ Over a third (35.7%) pointed to time pressure and tight deadlines, while 31% highlighted AI's role in research, data processing, and information synthesis. Professional concerns and ethical issues were raised by 29.4% of respondents, with some expressing strong opposition to AI use and others calling for clearer disclosure norms. A further 27% cited workflow facilitation and content scaling, framing AI as a tool to meet high output demands. Image generation and writing and editing support were each cited by 22.2% of respondents: the former noted AI as a practical substitute for costly or unavailable stock photography, while the latter covered tasks such as grammar checking, editing, and drafting. Finally, 12.7% pointed to transparency and ethical frameworks, reflecting awareness of emerging disclosure norms and the need for editorial guidelines to govern AI adoption in the newsroom.

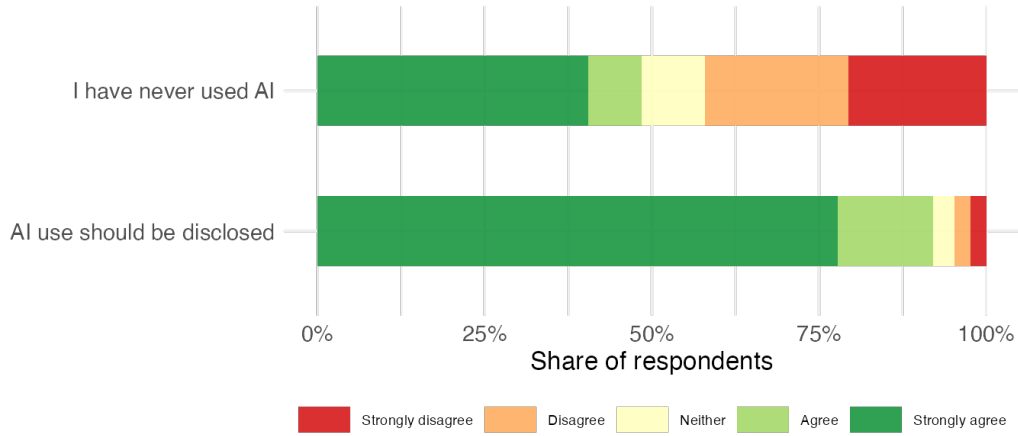
¹³Open-text responses were coded inductively using a two-step procedure: a language model first assigned granular topic labels to each response, then collapsed them into broad thematic categories. Categories are not mutually exclusive; respondents could mention multiple themes. Percentages reflect the share of the 126 respondents who provided a non-empty answer.

Figure 9: Journalists' perceptions

(a) Beliefs about Newsletter Sign-up by Treatment

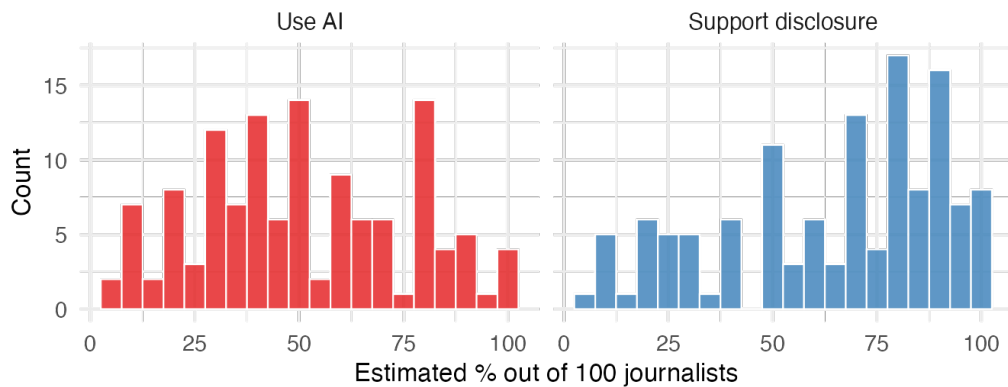


(b) Attitudes toward Personal Use of AI and its Disclosure



(c) Second-order Beliefs about AI Use and Disclosure Norms among Peers

Out of 100 people in your profession, how many you think...



Notes: Panel (a) displays mean beliefs about newsletter sign-up rates under each image condition, benchmarked against the actual Experiment 2 results. Panel (b) shows the distribution of journalists' personal attitudes toward AI use and disclosure. Panel (c) presents journalists' second-order beliefs about the prevalence of AI use and support for its disclosure among peers. $N = 132$.

7 Conclusion

We study whether AI-generated images attached to a real news article alter trust and engagement when text and outlet are held constant. Three findings stand out. First, on average, neither showing an AI image nor adding a label changes newsletter demand relative to a wire-service photograph. Second, effects hinge on what audiences *believe*: when respondents do not realise the image is synthetic, AI imagery shows higher engagement; when they do recognise it, engagement is comparably lower. Third, outlet perceptions move in parallel. AI imagery is linked to higher perceived quality, accuracy, and trustworthiness among respondents who think the image is real, and lower for all three among those who think it is AI. Labels temper this penalty only marginally. Results reconcile an aggregate behavioural null with offsetting belief updates, consistent with news-demand models in which credibility is a proximate driver of engagement (Chopra et al., 2024).

Our initial design is intentionally narrow to isolate this mechanism: one reputable outlet, one article on a politically salient but geographically distant event, compositionally matched imagery, and short-run outcomes in a UK online sample. Nonetheless, core findings replicate across six diverse articles and twelve AI-generated images in Experiment 2: across all article-image pairs, newsletter demand varies by no more than 2.4 p.p. across treatment arms, with none significantly different from control. External validity may vary with outlet reputation, topic polarity, image realism, audience priors, and the salience or design of disclosures. These limitations point to concrete next steps: newsroom or platform field experiments; variation in label design and placement; interventions that make provenance salient *ex ante* (e.g., prebunking or interactive cues); and measurement of revealed engagement (clicks, unsubscribes) across outlets and topics, including political advertising (Caprini, 2024; Novak, 2023; Park et al., 2024).

Experiment 2 also addresses the key identification challenge in Experiment 1: as recognition of the image source is elicited after the newsletter decision, the heterogeneity by recognition could, in principle, reflect selection rather than a causal belief mechanism. To isolate causality, we randomly vary the photorealism of the AI-generated images. Respondents assigned a less photorealistic AI image are significantly more likely to identify it as synthetic than those assigned a highly photorealistic version, while the depicted scene is held constant across both. Using the exogenous variation in recognition due to treatment assignment, we find that marginally rising the probability of correctly recognizing the image as AI leaves newsletter demand unchanged (2SLS coefficient: +0.015, s.e. 0.050, $p > 0.10$), consistent with the aggregate null arising from precisely offsetting belief-driven responses rather than from an absence of any behavioural mechanism, while worsening perceived outlet quality by 0.345 standard deviations ($p < 0.001$). These estimates are consistent in sign and magnitude with the OLS heterogeneity in Experiment 1, supporting the interpretation that belief updating about image origin — rather than the image itself or its label — is the operative mechanism driving both engagement and credibility responses.

Finally, to assess how industry practitioners perceive these dynamics, we surveyed 132 international journalists recruited via Prolific. Eliciting their beliefs about how AI imagery affects audience demand (anchored to the baseline newsletter sign-up rate from Experiment 2). We find that journalists are systematically miscalibrated in two directions. First, they substantially overestimate baseline newsletter demand, believing approximately 52 out of 100 readers would sign up compared to the actual rate of 35 observed in Experiment 2. Second, they anticipate that AI-generated images meaningfully depress demand — expecting sign-up rates of 38.5 for hard-to-recognize AI images and 24.4 for easy-to-recognize ones — a gradient of nearly 14 p.p. that finds no support in the data. At the same time, journalists display near-unanimous personal support for disclosure (91.1% in favour), yet underestimate their peers’ support (around 50 out of 100) while believing AI adoption among colleagues is considerably more widespread than their own self-reported behaviour implies. This disconnect between practitioner beliefs and empirical responses runs in both directions: journalists overestimate how much readers notice and penalise AI imagery, while simultaneously underestimating how widespread pro-disclosure norms already are among their peers, suggesting that the reputational risks of AI visual adoption may not be fully internalised in current newsroom decision-making.

For practitioners, the implication is pragmatic: if AI imagery is used to illustrate factual reporting by a trusted outlet, average engagement may look unchanged — but only so long as audiences do not notice the synthetic origin. Once they do, belief-based penalties emerge, and disclosure may soften credibility costs only at the margin. More broadly, aligning visual production with audience expectations in an era of synthetic media will require pairing transparency with strategies that sustain trust under uncertainty.

Data availability

Anonymized survey data from human participants, along with the replication code for this study, can be found at the following Github repository: [link](#).

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AI Images, Trust and News Demand

Maja Adena Eleonora Alabrese Francesco Capozza Isabelle Leader

A Theoretical Framework

This section adapts the Bayesian news-demand framework in [Chopra et al. \(2022\)](#) to our setting, where the informative object is not "this newsletter is fact-checked" but "this article seems to use an AI-generated image, possibly with a label." The key departure is that the informative signal is *endogenously noisy*: even an authentic photo is frequently believed to be AI (42.9–47.2% in the two control arms), and labels improve recognition only when the image is actually synthetic (51.6% in the AI+label arm). These patterns are documented in Figure 3a and Table B.3.

The model yields four testable implications that line up with the experimental results: (i) a **recognition penalty**: conditional on thinking the image is AI, perceived outlet quality and newsletter demand fall by about 10 p.p.; (ii) a **hidden-AI bonus**: conditional on *not* thinking the image is AI, AI imagery raises perceived outlet quality and demand by a similar amount (Figure C.1, Table B.5 and B.6); (iii) **weak label effects**: labels move recognition only slightly, so they do not close the demand differences between respondents who correctly and incorrectly identified the image source (Table B.5); and (iv) an **aggregate null**: because recognition is noisy for both real and AI images, the flows into the "high-demand" and "low-demand" posteriors almost cancel, so average newsletter demand stays in the narrow 36.5–39.3% band reported in Figure 4a.

A.1 Primitives

Outlet type The outlet has an unobservable editorial type

$$\theta \in \{H, L\},$$

where H denotes high editorial standards (accurate, trustworthy, professionally produced content) and L denotes lower standards. The reader holds a prior

$$\pi := () \in (0, 1),$$

which is common knowledge and does not itself depend on the image condition. It reflects the reader’s background belief about the outlet before exposure to any article.

Technology menu In each production period the outlet chooses a technology $a \in \{\text{AI}, \text{real}\}$ for the article image. Technologies differ along two dimensions: variable cost and recognisability.

Definition A.1 (Technology pairs). Let κ_R denote the variable cost of a real wire-service photograph and κ_{AI} the cost of an AI-generated image, with $\kappa_{\text{AI}} < \kappa_R$. The cost saving is $\delta := \kappa_R - \kappa_{\text{AI}} > 0$. Let $\rho \in [\rho^{\min}, \rho^{\max}]$ be a *photorealism parameter* chosen jointly with a : higher ρ (more realistic AI) lowers the probability that the reader classifies the image as synthetic.

The photorealism parameter ρ captures the Experiment 2 manipulation, where hard-to-detect and easy-to-detect AI images are randomly assigned.

Disclosure regime The outlet additionally chooses a binary disclosure label $\ell \in \{0, 1\}$, where $\ell = 1$ means a source cue is displayed (a wire-service credit for real photos; an AI-disclosure tag for synthetic ones). The pair (a, ℓ) fully characterises a treatment arm.

A.2 The Reader’s Information Problem

Image as a noisy signal After reading the article the reader observes the image but cannot directly observe (a, ℓ) . Instead, she forms a binary recognition judgment

$$r \in \{0, 1\}, \quad r = 1 \Leftrightarrow \text{“this looks AI-generated to me.”}$$

Conditional on the outlet’s type and the label status, recognition follows

$$(r = 1 \mid \theta, \ell) = p_{\theta\ell}, \quad \theta \in \{H, L\}, \ell \in \{0, 1\}.$$

[Diagnosticity] For both label conditions,

$$p_{L\ell} > p_{H\ell}.$$

The intuition for Assumption A.2 is that a low-standard outlet is more likely to put the reader in a situation where the image reads as synthetic, either because it actually used AI, used it carelessly, or provided weak contextual cues. Seeing $r = 1$ is therefore relatively more likely under .

[Label effects] Labels shift recognition in opposite directions depending on origin:

$$p_{H,1} < p_{H,0} \approx p_{L,0} < p_{L,1}.$$

That is, a wire-service credit slightly reduces false AI attributions for real photos, while an AI disclosure slightly increases correct detections for synthetic ones. The shifts are modest—consistent with the empirical finding that adding a label moves recognition rates by only four to six percentage points (?).

Bayesian belief updating Given the signal (r, ℓ) , the reader updates her belief about the outlet's type using Bayes' rule:

$$\mu(1, \ell) := (r = 1, \ell) = \frac{\pi p_{H\ell}}{\pi p_{H\ell} + (1 - \pi) p_{L\ell}}, \quad (1)$$

$$\mu(0, \ell) := (r = 0, \ell) = \frac{\pi(1 - p_{H\ell})}{\pi(1 - p_{H\ell}) + (1 - \pi)(1 - p_{L\ell})}. \quad (2)$$

Lemma A.2 (Recognition is bad news). *Under Assumption A.2,*

$$\mu(1, \ell) < \pi < \mu(0, \ell) \quad \text{for } \ell \in \{0, 1\}.$$

Proof. For $r = 1$, Assumption A.2 implies $p_{L\ell} > p_{H\ell}$, so the likelihood ratio

$$\frac{(r = 1 | \ell)}{(r = 0 | \ell)} = \frac{p_{H\ell}}{p_{L\ell}} < 1,$$

which, by Bayes' rule, pushes the posterior below the prior: $\mu(1, \ell) < \pi$. The $r = 0$ direction follows symmetrically using $1 - p_{L\ell} < 1 - p_{H\ell}$. $\square$

Lemma A.2 is the core insight: the reader rationally punishes the outlet whenever the image reads as AI-generated, because only a sloppier outlet would consistently produce that impression—regardless of whether the image is actually synthetic. Even readers exposed to an authentic wire-service photo penalise the outlet whenever they mistakenly classify it as AI.

A.3 Demand Decision

Payoffs The reader has an outside option normalised to zero. Subscribing to the newsletter yields expected utility

$$U(d = 1 | r, \ell) = v_H + (1 - \pi) v_L - c,$$

where $v_H > v_L \geq 0$ are the values of future newsletters from an H - and L -type outlet respectively, and $c \geq 0$ is an idiosyncratic subscription cost drawn from a continuous CDF F on $\mathbb{R}_+$, independent of all signals. The cost c captures the heterogeneous opportunity cost of inbox attention (time, effort, signal-to-noise preferences) that generates variation in sign-up rates across readers.

Subscription rule The reader subscribes if and only if $c \leq (r, \ell)$, where

$$(r, \ell) := (v_H - v_L) + v_L \quad (3)$$

is the willingness-to-pay threshold. Since is strictly increasing in and F is non-decreasing with positive density, we define the demand probability conditional on signal (r, ℓ) as

$$D(r, \ell) := F((r, \ell)). \quad (4)$$

Proposition A.3 (Recognition penalty in demand). *For each label condition $\ell \in \{0, 1\}$,*

$$D(0, \ell) > D(1, \ell).$$

Proof. Lemma A.2 gives $\mu(0, \ell) > \mu(1, \ell)$. Since is strictly increasing in μ (equation (3)) and F has positive density in a neighbourhood of , it follows that $D(0, \ell) = F((0, \ell)) > F((1, \ell)) = D(1, \ell)$. $\square$

Proposition A.3 corresponds to the ~ 10 percentage-point demand gap documented in both experiments: readers who correctly identify AI images subscribe less; readers who do not subscribe more.

A.4 From Signals to Treatment Arms

Within treatment arm $t \in \{\text{unlabelled real, labelled real, unlabelled AI, labelled AI}\}$, the fraction of readers reporting $r = 1$ is the arm-specific attribution rate α_t . Average demand in arm t is therefore a mixture:

$$_t = \alpha_t D(1, \ell_t) + (1 - \alpha_t) D(0, \ell_t), \quad (5)$$

where $\ell_t \in \{0, 1\}$ is the label status of arm t .

Proposition A.4 (Aggregate null). *Suppose $\alpha_t \in [\underline{\alpha}, \bar{\alpha}]$ with $\bar{\alpha} - \underline{\alpha}$ small, and that the demand gap $D(0, \ell) - D(1, \ell)$ is approximately equal across arms. Then*

$$_t \approx \alpha D(1, \ell) + (1 - \alpha) D(0, \ell) \quad \forall t,$$

and average demand is nearly constant across treatment arms.

Proof sketch. Substituting the empirical attribution rates $\alpha_t \in \{0.472, 0.429, 0.458, 0.516\}$ Table B.2 into (5), and using the fact that $D(0, \ell) - D(1, \ell)$ is approximately constant across arms (Proposition A.3), the weighted sum $_t$ varies by at most $(\bar{\alpha} - \underline{\alpha}) \cdot [D(0) - D(1)]$. With $\bar{\alpha} - \underline{\alpha} \approx 0.09$ and a demand gap of 0.10, the maximum spread across arms is ≈ 0.009 , consistent with the narrow 36.5–39.3% band observed in Experiment 1. $\square$

Remark A.5. Proposition A.4 rationalises the aggregate null without assuming that AI images are irrelevant to readers. The null arises from symmetric cancellation: every arm puts roughly half the sample into the high-demand state ($r = 0$) and half into the low-demand state ($r = 1$). The mechanism is analogous to Chopra et al. (2024), where fact-checking raises perceived accuracy among some readers while confirming priors for others, leaving average demand approximately unchanged.

A.5 The Outlet's Technology Choice

Per-period profit In each period the outlet maximises profit Π over the technology choice (a, ρ, ℓ) :

$$\Pi(a, \rho, \ell) = (a, \rho, \ell) \cdot m - \kappa(a) - \phi(\ell), \quad (6)$$

where $m > 0$ is the margin per subscriber (advertising or subscription revenue), $\kappa(a)$ is the variable image-production cost, and $\phi(\ell) \geq 0$ is the disclosure cost (legal, operational, or reputational). Since Proposition A.4 implies is approximately invariant across arms, the outlet always weakly prefers AI adoption on direct cost grounds:

$$\Pi(\text{AI}, \rho, \ell) - \Pi(\text{real}, \rho, \ell) \approx \delta - \Delta\phi(\ell) \geq 0 \quad \text{when } \delta \geq \Delta\phi(\ell).$$

Reputational cost of recognition This comparison holds only for *average* demand. Define the recognition rate under AI adoption as $\alpha_{\text{AI}}(\rho, \ell)$, which is decreasing in photorealism ρ and increasing in label presence ℓ (by Assumption A.2). The demand shortfall from recognised AI, relative to the real-photo counterfactual, is

$$\Delta D^{\text{rep}} = \alpha_{\text{AI}}(\rho, \ell) [D(0, \ell_R) - D(1, \ell_{\text{AI}})], \quad (7)$$

where the bracketed term is the demand gap guaranteed by Proposition A.3.

The outlet therefore faces a cost–recognition trade-off. Switching to AI reduces κ but inflates α_{AI} and erodes demand among recognising readers. The photorealism parameter ρ is the internal instrument: increasing ρ lowers recognition (and thus the reputational cost) at the expense of higher AI generation quality costs. The interior optimum satisfies

$$\frac{\partial \kappa_{\text{AI}}}{\partial \rho} = -m \cdot \frac{\partial \alpha_{\text{AI}}}{\partial \rho} \cdot [D(0, \ell) - D(1, \ell)], \quad (8)$$

equating the marginal cost of higher image quality to the marginal demand gain from reduced recognition.

A.6 The Role of Disclosure

Label as a belief shifter A disclosure label $\ell = 1$ operates through two channels. The *direct channel* raises $p_{L,1}$ (AI images are more often correctly identified when labelled) and lowers $p_{H,1}$ (real photos are somewhat less likely to be mistaken for AI). The *indirect channel* shifts the mixture weights in τ via equation (5) but does not change the height of the gap $D(0, \ell) - D(1, \ell)$.

Proposition A.6 (Label cannot close the recognition gap). *Under Assumption A.2 (modest label effects), even with $\ell = 1$,*

$$D(0, \ell) - D(1, \ell) > 0$$

remains economically large. Disclosure tempers but does not remove the recognition penalty.

Proof. The gap $D(0, \ell) - D(1, \ell) = F((0, \ell)) - F((1, \ell))$ depends on $\mu(0, \ell) - \mu(1, \ell)$, which in turn depends on $p_{\theta\ell}$ through equations (1)–(2). Assumption A.2 restricts labels to shift $p_{\theta\ell}$ by at most a few percentage points. For the gap to close entirely, we would need $\mu(0, \ell) \rightarrow \mu(1, \ell)$, which requires either $p_{H\ell} \rightarrow p_{L\ell}$ (perfect recognition) or F to be degenerate at a common threshold. Neither condition holds in the data. $\square$

A.7 Transparency bonus for realistic AI imagery

A nuanced result emerges for hard-to-detect AI images combined with a label. When the AI image is photorealistic (ρ high), the baseline recognition rate α_{AI} is low. In this case, the label serves a different function: it preempts the inference “this outlet uses AI carelessly” and replaces it with “this outlet is transparent about AI use.” Formally, among readers who would have remained sceptical without the label, the label shifts the prior on the outlet’s type upward, partially offsetting the recognition penalty. This is consistent with the labelled hard-AI arm of Experiment 2, where the demand penalty among correctly recognising readers is near zero (?).

A.8 Equilibrium Characterisation

The model has the following sequential structure with no backward induction loop:

1. **Nature** draws $\theta \in \{H, L\}$ with probability $(\pi, 1 - \pi)$.
2. **Outlet** observes θ and chooses (a, ρ, ℓ) to maximise $\Pi(a, \rho, \ell)$.
3. **Reader** observes (image, ℓ), forms recognition judgment r , updates to τ , draws private cost $c \sim F$, and subscribes iff $c \leq (r, \ell)$.

Because θ is drawn before the outlet acts and the reader does not observe θ directly, the model is a Bayesian signalling structure. Crucially, the reader’s demand decision does not feed

back into θ within the period: quality is fixed, not endogenously sustained by readership. This removes the circularity present in Klein–Leffler reputation cycles.

Comparative statics Table A.1 summarises the key comparative statics of the model.

Table A.1: Comparative statics

Parameter change	Effect on t	Effect on AI adoption incentive
Higher photorealism $\rho \uparrow$	Ambiguous (shifts α_t down, raises $D(0, \ell)$ path weight)	Increases AI adoption
Larger cost saving $\delta \uparrow$	None (demand approx. invariant)	Increases AI adoption
Stronger label effects (higher literacy)	Small increase for AI arms; small decrease for real arms	Reduces AI adoption if reputational cost becomes salient
Higher reader prior $\pi \uparrow$	Demand rises uniformly; gap $D(0, \ell) - D(1, \ell)$ shrinks	Reduces reputational cost of recognition
Larger value spread $v_H - v_L \uparrow$	Gap $D(0, \ell) - D(1, \ell)$ widens	Increases reputational cost of recognition

A.9 Extension: Heterogeneous Reader Types

The baseline model treats all readers symmetrically. A natural extension stratifies readers by AI literacy $\lambda \in [0, 1]$, where λ captures the probability that a reader correctly identifies the image type regardless of label. Under this extension:

- High- λ readers always recover r correctly, so their beliefs coincide with the true image origin.
- Low- λ readers behave as in the baseline: their r is noisy and governed by $p_{\theta\ell}$.

Aggregate demand becomes

$$D_t^\lambda = \lambda D(r^*, \ell) + (1 - \lambda) D_t, \quad (9)$$

where r^* is the correct recognition outcome for that arm.

Corollary A.7 (Dissolution of the aggregate null). *As the population becomes more AI-literate ($\lambda \rightarrow 1$), the aggregate null dissolves: recognising AI imagery correctly now consistently reduces demand, eliminating the offsetting cancellation mechanism.*

Corollary A.7 predicts that, as AI detection tools diffuse and media-literacy interventions take effect, the empirical null will shift toward a monotone negative average effect of AI adoption on demand. This is a testable comparative static for future work.

A.10 Key Predictions

The framework generates five testable predictions, all consistent with the empirical findings.

1. **Aggregate null.** Average newsletter demand is approximately constant across treatment arms, because recognition rates cluster near 50% in every arm, generating offsetting positive and negative demand shifts that cancel in the population mixture.
2. **Belief-driven heterogeneity.** Conditional on believing the image is real ($r = 0$), demand is uniformly higher; conditional on believing it is AI ($r = 1$), demand is uniformly lower—regardless of whether that belief is correct.
3. **Label modesty.** Labels move average demand little: they shift recognition rates only modestly (Assumption A.2), and the demand gap $D(0, \ell) - D(1, \ell)$ is determined by the prior π and the value spread $v_H - v_L$, not by the label per se.
4. **Transparency bonus.** For hard-to-detect AI images, labelling can partially protect demand among recognising readers by reframing the inference from “careless AI use” to “transparent AI use.” This is a second-order effect and depends on reader sophistication.
5. **Reputational cost of recognised AI.** Even when the AI image is of objectively high quality (i.e., the outlet’s true type is H), recognition of AI use triggers a downward belief revision, because readers cannot observe θ and update on the signal $r = 1$ as if AI use correlates with lower standards. The outlet bears a reputational cost that is independent of actual quality.

B Appendix Tables

Table B.1: Summary Statistics: Experiment 1-2 vs. UK population

Variable (definition)	Experiment 1	Experiment 2	UK Population
Gender (share female)	52.3%	51.5%	51.1%
Age (years)	46.7	44.4	40.7 (median)
Ethnicity (share White)	88.3%	87.3%	83.0%
UK-born (share born in the UK)	89.1%	93.5.3%	84.0%
Region (share living in the South East)	14.8%	16.1%	15.6%
Income (share earning > £40K)	41.7%	59.2%	25.0%
Employment (share employed)	75.8%	76.1%	75.1%
Education (share with university degree)	62.3%	64.0%	33.8%
Party (share previously voting Labour)	36.6%	27.1%	33.7%

Notes: *Experiment 1* and *Experiment 2* figures refer to our survey respondents in the two waves, respectively. *UK Population* benchmarks draw on the 2021–2022 UK Census (England, Wales, Scotland, Northern Ireland) where possible. When no harmonised UK-wide statistic was available (e.g., income, education, regional population shares), benchmark values are based on the closest nationally representative source. Gender: female share in UK mid-2022; Age: UK median age (ONS); Ethnicity: White share; UK-born: share born in the UK; Region: population in South East; Income: individuals earning above the 75th percentile (HMRC SPI); Employment: employment rate among ages 16–64; Education: share with Level 4+ (England & Wales); Party: Labour vote share in the July 2024 UK General Election.

Table B.2: Balance across Treatments: Group Means, Global and Pairwise Tests

Variable	Unlabeled Control	Labeled Control	Unlabeled AI	Labeled AI	Global_p	Min_pairwise_p
Gender (share of 'Female')	0.54	0.52	0.51	0.53	0.31	1.00
Age	46.36	46.61	47.60	46.20	0.31	0.51
Ethnicity (share of 'White')	0.87	0.89	0.88	0.89	0.17	1.00
UKborn (share of 'Yes')	0.89	0.89	0.89	0.89	0.80	1.00
Region (share of 'South East')	0.15	0.13	0.15	0.16	0.37	0.90
Income (share of '£20,000 - £29,999')	0.15	0.17	0.18	0.17	0.52	0.81
Employment (share of 'Employed full time')	0.51	0.54	0.52	0.53	0.79	1.00
Education (share of 'University or higher')	0.62	0.62	0.62	0.63	0.78	1.00
Political (share of 'Liberal')	0.40	0.40	0.42	0.37	0.22	0.43
Party (share of 'Labour')	0.37	0.36	0.37	0.37	0.96	1.00
InterestNews (share of 'Interested')	0.45	0.45	0.42	0.40	0.42	0.26
ConsumptionNews (share of 'Every day')	0.56	0.59	0.58	0.56	0.79	1.00
ArticleTime (seconds reading article)	93.64	97.46	100.07	103.76	0.18	0.20
Post-treatment variables						
Manipulation (share believe AI)	0.47	0.43	0.46	0.52	0.01	0.01
OriginCorrect (share identify AI/Not)	0.53	0.57	0.46	0.52	0.00	0.00

Notes: This table reports the mean of each demographic and attitudinal characteristic by treatment arm. For continuous variables, the reported p -values come from an ANOVA test of equality of means across treatment groups. For categorical variables, p -values correspond to Pearson χ^2 tests of independence. The column *Min pairwise p* shows the smallest p -value from pairwise t - or proportion tests between any two treatment arms.

Table B.3: Manipulation Check

Manipulation ($AI = Yes$)				
	(1) OLS	(2) OLS	(3) Logit	(4) Logit
Constant	0.471*** (0.019)	7.82×10^{-13} (8.02×10^{-13})	-0.114 (0.075)	-14.6*** (1.57×10^{-5})
AI	-0.013 (0.026)	-0.025 (0.027)	-0.054 (0.106)	-0.107 (0.116)
Label	-0.042 (0.026)	-0.042 (0.026)	-0.170 (0.106)	-0.184 (0.114)
AI $\times$ Label	0.100*** (0.037)	0.109*** (0.038)	0.403*** (0.150)	0.472*** (0.164)
Pseudo R^2	0.00272	0.05798	0.00284	0.06158
Observations	2,870	2,870	2,870	2,870
Controls		x		x
Mean DV	0.469	0.469	0.469	0.469

Notes: *Manipulation* equals 1 if respondents said the item was AI-generated, 0 otherwise. *AI* equals 1 for treatment arms Unlabeled AI (UAI) and Labeled AI (LAI), 0 for control arms (with or without labels). *Label* equals 1 for labeled control and LAI (label present), 0 otherwise. Controls include demographics, political orientation, and media familiarity. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Table B.4: AI and Label Effects on Requesting the Newsletter

	Demand (<i>Newsletter = Yes</i>)							
	(1) OLS	(2) OLS	(3) Logit	(4) Logit	(5) OLS	(6) OLS	(7) Logit	(8) Logit
Constant	0.364*** (0.018)	6.73×10^{-13} (6.9×10^{-13})	-0.556*** (0.077)	-14.6*** (1.01×10^{-5})	0.313*** (0.025)	-0.082** (0.035)	-0.788*** (0.117)	-15.0*** (0.176)
AI	0.006 (0.025)	-0.0007 (0.025)	0.025 (0.110)	-0.005 (0.122)	0.105*** (0.036)	0.083** (0.034)	0.455*** (0.156)	0.413** (0.172)
Label	0.029 (0.026)	0.029 (0.025)	0.123 (0.109)	0.143 (0.121)	0.045 (0.037)	0.036 (0.036)	0.204 (0.167)	0.191 (0.183)
AI $\times$ Label	-0.027 (0.036)	-0.009 (0.035)	-0.116 (0.154)	-0.049 (0.174)	-0.051 (0.052)	-0.017 (0.051)	-0.228 (0.225)	-0.107 (0.252)
OriginCorrect					0.098*** (0.036)	0.082** (0.035)	0.426*** (0.157)	0.402** (0.176)
AI $\times$ OriginCorrect					-0.201*** (0.051)	-0.171*** (0.050)	-0.874*** (0.222)	-0.861*** (0.249)
Label $\times$ OriginCorrect					-0.036 (0.051)	-0.019 (0.050)	-0.165 (0.221)	-0.103 (0.245)
AI $\times$ Label $\times$ OriginCorrect					0.062 (0.072)	0.031 (0.070)	0.282 (0.313)	0.191 (0.352)
Pseudo R ²	0.00037	0.10309	0.00039	0.10844	0.00615	0.10813	0.00646	0.11368
Observations	2,870	2,870	2,870	2,870	2,870	2,870	2,870	2,870
Controls		x		x		x		x
Mean DV	0.375	0.375	0.375	0.375	0.375	0.375	0.375	0.375

Notes: *Demand* equals 1 if respondents said they wish to receive the Newsletter, 0 otherwise. *AI* equals 1 for treatment arms Unlabeled AI (UAI) and Labeled AI (LAI), 0 for control arms (with or without labels). *Label* equals 1 for labeled control and LAI (label present), 0 otherwise. *OriginCorrect* equals 1 if respondents correctly recognized whether the image was AI-generated or not, 0 otherwise. Controls include demographics, political orientation, and media familiarity. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Table B.5: AI and Label Effects on Requesting the Newsletter

Demand (<i>Newsletter = Yes</i>)								
	<i>OriginCorrect</i> = 1				<i>OriginCorrect</i> = 0			
	(1) OLS	(2) OLS	(3) Logit	(4) Logit	(5) OLS	(6) OLS	(7) Logit	(8) Logit
Constant	0.411*** (0.025)	-3.72×10^{-13} (3.91×10^{-13})	-0.362*** (0.104)	-14.6*** (4.55×10^{-6})	0.313*** (0.025)	1.16*** (0.203)	-0.788*** (0.117)	16.4*** (1.14)
AI	-0.097*** (0.036)	-0.090** (0.037)	-0.420*** (0.158)	-0.456** (0.193)	0.105*** (0.036)	0.081** (0.035)	0.455*** (0.156)	0.466** (0.190)
Label	0.009 (0.035)	0.013 (0.035)	0.039 (0.145)	0.067 (0.173)	0.045 (0.037)	0.037 (0.037)	0.204 (0.167)	0.231 (0.202)
AI $\times$ Label	0.011 (0.050)	0.010 (0.050)	0.054 (0.218)	0.070 (0.262)	-0.051 (0.052)	-0.009 (0.053)	-0.228 (0.225)	-0.102 (0.282)
Pseudo R ²	0.00654	0.12896	0.00686	0.13710	0.00571	0.16164	0.00602	0.17029
Observations	1,488	1,488	1,488	1,488	1,382	1,382	1,382	1,382
Controls		x		x		x		x
Mean DV	0.375	0.375	0.375	0.375	0.375	0.375	0.375	0.375

Notes: *Demand* equals 1 if respondents said they wish to receive the Newsletter, 0 otherwise. *AI* equals 1 for treatment arms Unlabeled AI (UAI) and Labeled AI (LAI), 0 for control arms (with or without labels). *Label* equals 1 for labeled control and LAI (label present), 0 otherwise. *OriginCorrect* equals 1 if respondents correctly recognized whether the image was AI-generated or not, 0 otherwise. Controls include demographics, political orientation, and media familiarity. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Table B.6: Mechanisms: Effects of AI and Label on Perceived Outlet Quality

	NotAccurate		NotTrustworthy		LowQuality		Accurate		Trustworthy		Quality		z_index	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
Constant	0.186*** (0.021)	0.143*** (0.023)	0.233*** (0.023)	0.172*** (0.025)	0.198*** (0.022)	0.138*** (0.025)	3.03*** (0.041)	3.15*** (0.463)	2.92*** (0.040)	1.69*** (0.305)	3.06*** (0.042)	4.54*** (0.660)	-0.376*** (0.049)	-0.234 (0.518)
AI	-0.165*** (0.022)	-0.157*** (0.022)	-0.200*** (0.025)	-0.187*** (0.024)	-0.159*** (0.024)	-0.157*** (0.024)	0.602*** (0.052)	0.536*** (0.049)	0.543*** (0.052)	0.476*** (0.049)	0.526*** (0.055)	0.480*** (0.054)	0.739*** (0.062)	0.660*** (0.058)
Label	-0.037 (0.029)	-0.041 (0.029)	-0.023 (0.033)	-0.020 (0.032)	-0.040 (0.030)	-0.041 (0.030)	0.068 (0.056)	0.067 (0.053)	0.025 (0.056)	0.004 (0.054)	0.119** (0.060)	0.096 (0.059)	0.093 (0.067)	0.073 (0.064)
OriginCorrect	-0.154*** (0.023)	-0.143*** (0.023)	-0.188*** (0.025)	-0.172*** (0.025)	-0.142*** (0.025)	-0.138*** (0.025)	0.585*** (0.054)	0.514*** (0.051)	0.586*** (0.054)	0.520*** (0.052)	0.511*** (0.056)	0.458*** (0.055)	0.744*** (0.064)	0.660*** (0.061)
AI × Label	0.034 (0.031)	0.042 (0.031)	0.025 (0.035)	0.031 (0.035)	0.044 (0.033)	0.047 (0.034)	-0.206*** (0.074)	-0.199*** (0.071)	-0.121 (0.074)	-0.096 (0.071)	-0.187** (0.078)	-0.143* (0.078)	-0.227*** (0.087)	-0.194** (0.084)
AI × OriginCorrect	0.329*** (0.033)	0.304*** (0.032)	0.402*** (0.036)	0.372*** (0.035)	0.335*** (0.035)	0.323*** (0.035)	-1.20*** (0.075)	-1.06*** (0.072)	-1.14*** (0.076)	-1.01*** (0.073)	-1.05*** (0.080)	-0.959*** (0.079)	-1.50*** (0.090)	-1.34*** (0.086)
Label × OriginCorrect	0.025 (0.031)	0.029 (0.032)	0.013 (0.035)	0.008 (0.035)	0.026 (0.034)	0.018 (0.034)	-0.115 (0.074)	-0.117* (0.071)	-0.093 (0.074)	-0.074 (0.071)	-0.140* (0.078)	-0.114 (0.077)	-0.153* (0.088)	-0.135 (0.084)
AI × Label × OriginCorrect	-0.019 (0.045)	-0.024 (0.045)	-0.019 (0.050)	-0.020 (0.049)	-0.045 (0.049)	-0.041 (0.048)	0.253** (0.104)	0.221** (0.100)	0.200* (0.105)	0.153 (0.101)	0.196* (0.111)	0.127 (0.109)	0.287** (0.124)	0.222* (0.119)
Adjusted R ²	0.07289	0.10212	0.08450	0.14188	0.06044	0.09112	0.13197	0.23147	0.12181	0.22221	0.09699	0.16363	0.14587	0.25182
Observations	2,870	2,870	2,870	2,870	2,870	2,870	2,869	2,869	2,869	2,869	2,869	2,869	2,869	2,869
Controls	x		x		x		x		x		x		x	
Mean DV	0.098	0.098	0.129	0.129	0.118	0.118	3.33	3.33	3.20	3.20	3.33	3.33	0	0

Notes: Each column reports OLS estimates of treatment effects on the perceived outlet quality measures. Dependent variables include *Accuracy*, *Trustworthiness*, and *Quality* (measured on 1–5 Likert scales), their binary counterparts (we create a dummy equal to 1 if respondents selected one of the two lowest categories in the Likert scale, 0 otherwise), and standardized versions of these measures (z-scores) averaged into a composite perceived outlet quality *z_index*. *AI* equals 1 for treatment arms Unlabeled AI (UAI) and Labeled AI (LAI), 0 for control arms (with or without labels). *Label* equals 1 for labeled control and LAI (label present), 0 otherwise. *OriginCorrect* equals 1 if respondents correctly recognized whether the image was AI-generated or not, 0 otherwise. Controls include demographics, political orientation, and media familiarity. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Table B.7: AI and Label Effects on Requesting the Newsletter and on Perceived Outlet Quality (IV estimates)

	First stage		Second stage			
	Manipulation		Demand IV		z_index	
AI	-0.013 (0.026)	-0.025 (0.027)	0.002 (0.023)	-0.003 (0.020)	-0.019 (0.042)	-0.026 (0.037)
Label	-0.042 (0.026)	-0.042 (0.026)	0.017 (0.019)	0.025 (0.018)	-0.025 (0.034)	-0.032 (0.033)
AI × Label	0.100*** (0.037)	0.109*** (0.038)				
Manipulation			-0.274 (0.366)	-0.084 (0.324)	-1.448** (0.675)	-1.317** (0.595)
Weak IV F	7.243	8.424				
Wu–Hausman p			0.6010	0.9856	0.2084	0.1957
Adj. R ²	0.003	0.035	-0.030	0.097	-0.039	0.103
Observations	2870	2870	2870	2870	2870	2870
Controls		x		x		x
Mean DV	0.469	0.469	0.375	0.375	0	0

Notes: Columns (1)–(2) report first stage estimates where *Manipulation* is regressed on *AI*, *Label*, and their interaction. Columns (3)–(4) report 2SLS estimates for *Demand*, an indicator for requesting the newsletter. Columns (5)–(6) report 2SLS estimates for *z_index*, an index of perceived outlet quality (higher = better) that averages standardized responses for *Accuracy*, *Trustworthiness*, and *Quality*, each originally measured on 1–5 Likert scales. In the 2SLS specification, *AI* × *Label* is used as the excluded instrument for *Manipulation*. *Manipulation* equals 1 if the respondent identified the image as AI-generated, 0 otherwise. *AI* equals 1 for Unlabeled AI (UAI) and Labeled AI (LAI), 0 for control arms. *Label* equals 1 for labeled control and LAI, 0 otherwise. *Weak-IV F* reports the robust partial *F*–statistic for the excluded instrument in the first stage, and *Wu–Hausman p* reports the *p*–value from a robust Durbin–Wu–Hausman exogeneity test for *Manipulation*. Controls include demographics, political orientation, and media familiarity. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Table B.8: Mechanisms (Alternative): Effects of AI and Label on Perceived Bias, Complexity, and Engagement

	Bias		Complex		Researchertrust		Entertaining	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Constant	-0.245*** (0.045)	0.758 (0.472)	2.65*** (0.043)	4.61*** (0.408)	3.13*** (0.042)	3.00*** (0.291)	2.40*** (0.044)	1.76*** (0.395)
AI	-0.052 (0.055)	-0.070 (0.056)	0.011 (0.057)	-0.009 (0.058)	0.414*** (0.053)	0.351*** (0.052)	0.194*** (0.059)	0.154*** (0.059)
Label	0.019 (0.065)	-0.017 (0.066)	0.012 (0.060)	0.007 (0.061)	0.070 (0.058)	0.042 (0.058)	0.089 (0.062)	0.086 (0.063)
OriginCorrect	-0.053 (0.056)	-0.063 (0.057)	0.034 (0.060)	0.035 (0.059)	0.359*** (0.057)	0.303*** (0.056)	0.166*** (0.061)	0.111* (0.059)
AI × Label	-0.073 (0.081)	-0.027 (0.082)	0.067 (0.082)	0.090 (0.083)	-0.179** (0.076)	-0.123 (0.076)	-0.206** (0.083)	-0.175** (0.085)
AI × OriginCorrect	0.097 (0.079)	0.107 (0.079)	-0.158* (0.083)	-0.133 (0.082)	-0.735*** (0.080)	-0.609*** (0.078)	-0.356*** (0.086)	-0.279*** (0.084)
Label × OriginCorrect	-0.036 (0.080)	0.0001 (0.081)	0.025 (0.081)	0.034 (0.081)	-0.046 (0.078)	-0.024 (0.076)	-0.093 (0.085)	-0.086 (0.083)
AI × Label × OriginCorrect	0.056 (0.112)	0.007 (0.112)	-0.117 (0.114)	-0.160 (0.115)	0.221** (0.110)	0.136 (0.109)	0.207* (0.119)	0.190 (0.116)
Adjusted R ²	0.00012	0.02806	0.00674	0.03025	0.04422	0.12409	0.00675	0.08805
Observations	2,869	2,869	2,869	2,869	2,869	2,869	2,869	2,869
Controls		x		x		x		x
Mean DV	-0.286	-0.286	2.649	2.649	3.347	3.347	2.492	2.492

Notes: Each column reports OLS estimates for alternative mechanisms, including *Perceived political bias*, *Complexity*, *Entertainment value*, and *Trust in researcher*. All dependent variables are continuous measures, coded on 1–5 Likert-type scales, except for *Bias*, which ranges from –2 (very left-wing) to +2 (very right-wing). *AI* equals 1 for treatment arms Unlabeled AI (UAI) and Labeled AI (LAI), 0 for control arms (with or without labels). *Label* equals 1 for labeled control and LAI (label present), 0 otherwise. *OriginCorrect* equals 1 if respondents correctly recognized whether the image was AI-generated or not, 0 otherwise. Controls include demographics, political orientation, and media familiarity. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Table B.9: Guessed Topic of Experiment

Category	Share	Example
AI / Artificial Intelligence	53.50	Whether humans can tell whether an article is generated by AI
Media / News / Social media	24.50	Fake news
Other / Unclear	14.60	No idea.
Politics / Elections / Opinion	4.30	Political awareness
Trust / Credibility / Ethics	2.30	Trust in governments
Psychology / Perception	0.40	attitudes towards immigration
Economics / Finance / Work	0.10	President Trump's decision to deploy 2000 National Guard workers to...
Marketing / Advertising	0.10	Persuasion
Misinformation / Fake news	0.10	To see who believes fake stories

Notes: Shares are percentages of non-empty responses to *"If you were to guess, what is this study about?"*. Categories are assigned via a simple keyword dictionary; the example quote is the first response observed in that category (truncated for space).

Table B.10: Reasons for Requesting (or Not Requesting) the Newsletter

Category	Share	Example
Curiosity / Novelty	47.00	Not interested, fed up of reading stuff about the US, they would no...
Interest in AI / Technology	33.50	The newsletter seemed written by a human and telling s story about ...
Other / Unclear	13.60	I enjoyed reading the article and it gave me a better insight into ...
Interest in Science / Research	3.40	The story presented seemed interesting and well researched, but I g...
Trust / Credibility	1.80	it seemed like it was on the side of truthful with all the informat...
Skepticism / Disinterest	0.60	I think it was quite a boring read, but I do not enjoy reading a la...
Time / Effort / Relevance	0.20	I found the article too long and drawn out. There was too much info...

Notes: Shares are percentages of non-empty responses to *"Could you please explain your decision regarding the newsletter?"*. Categories are assigned via a keyword dictionary; the example quote is the first response observed in that category (truncated).

Table B.11: Balance across Treatments: Group Means, Global and Pairwise Tests

Variable	Control	Control + Label	Easy AI	Easy AI + Label	Hard AI	Hard AI + Label	Global_p	Min_pairwise_p
Gender (share of 'Female')	0.54	0.51	0.52	0.51	0.51	0.5	0.52	1.00
Age	44.93	44.73	43.63	44.14	44.7	44.08	0.57	1.00
Ethnicity (share of 'White')	0.86	0.88	0.85	0.89	0.88	0.89	0.93	1.00
UKborn (share of 'Yes')	0.93	0.93	0.94	0.94	0.92	0.96	0.05	0.05
Region (share of 'South East')	0.17	0.15	0.16	0.14	0.16	0.18	0.98	1.00
Income (share of '£20,000 - £29,999')	0.17	0.16	0.15	0.16	0.17	0.15	0.64	1.00
Employment (share of 'Employed full time')	0.55	0.6	0.58	0.56	0.58	0.57	0.57	0.69
Education (share of 'University or higher')	0.62	0.66	0.65	0.66	0.62	0.64	0.49	1.00
Political (share of 'Liberal')	0.38	0.41	0.4	0.37	0.35	0.42	0.24	0.19
Party (share of 'Labour')	0.28	0.25	0.31	0.27	0.23	0.29	0.16	0.02
InterestNews (share of 'Interested')	0.43	0.41	0.4	0.39	0.37	0.38	0.60	0.42
ConsumptionNews (share of 'Every day')	0.55	0.57	0.53	0.56	0.54	0.54	0.94	1.00
Newsletter (share of 'Yes')	0.16	0.17	0.15	0.15	0.14	0.14	0.47	0.80
<i>Post-treatment variables</i>								
ArticleTime (seconds reading article)	80.01	76.05	80.25	75.49	80.05	73.69	0.78	1.00
Manipulation (share believe AI)	0.37	0.36	0.81	0.87	0.44	0.52	0.00	0.00
OriginCorrect (share identify AI/Not)	0.63	0.64	0.81	0.87	0.44	0.52	0.00	0.00

Notes: This table reports the mean of each demographic and attitudinal characteristic by treatment arm. For continuous variables, the reported p -values come from an ANOVA test of equality of means across treatment groups. For categorical variables, p -values correspond to Pearson χ^2 tests of independence. Categorical rows show the share of the modal category. The column *Min pairwise p* shows the smallest Holm-adjusted p -value from pairwise t - or proportion tests between any two treatment arms.

Table B.12: AI and Label Effects on Manipulation and Correct Recognition

	Manipulation (<i>AI = Yes</i>)				OriginCorrect (<i>= Yes</i>)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Label Control	-0.002 (0.026)	0.0010 (0.025)			0.007 (0.026)	0.006 (0.026)		
Easy AI	0.446*** (0.023)	0.451*** (0.023)	0.446*** (0.023)	0.451*** (0.023)	0.177*** (0.023)	0.174*** (0.023)	0.177*** (0.023)	0.174*** (0.023)
Easy AI + Label	0.496*** (0.022)	0.499*** (0.022)			0.232*** (0.022)	0.231*** (0.022)		
Hard AI	0.073*** (0.026)	0.072*** (0.025)	0.073*** (0.026)	0.072*** (0.025)	-0.194*** (0.026)	-0.192*** (0.026)	-0.194*** (0.026)	-0.192*** (0.026)
Hard AI + Label	0.154*** (0.026)	0.156*** (0.026)			-0.111*** (0.026)	-0.113*** (0.026)		
Label			-0.002 (0.026)	0.0010 (0.025)			0.007 (0.026)	0.006 (0.026)
Easy AI × Label			0.051 (0.032)	0.048 (0.032)			0.048 (0.032)	0.051 (0.032)
Hard AI × Label			0.083** (0.037)	0.084** (0.036)			0.076** (0.037)	0.073** (0.037)
Pseudo R ²	0.141	0.152	0.141	0.152	0.084	0.091	0.084	0.091
Observations	4,196	4,196	4,196	4,196	4,196	4,196	4,196	4,196
Image FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	Yes	No	Yes	No	Yes	No	Yes
Mean DV	0.561	0.561	0.561	0.561	0.652	0.652	0.652	0.652

Notes: *Manipulation* equals 1 if respondents identified the image as AI-generated, 0 otherwise. *OriginCorrect* equals 1 if respondents correctly recognized whether the image was AI-generated or not, 0 otherwise. *Easy AI* equals 1 for treatment arms Easy AI Unlabeled (EAI) and Easy AI Labeled (LEAI), 0 otherwise; *Hard AI* equals 1 for Hard AI Unlabeled (HAI) and Hard AI Labeled (LHAI), 0 otherwise; *Label* equals 1 if a label was present (Label Control, LEAI, LHAI), 0 otherwise. Controls include age, gender, education, income, political orientation, newsletter subscription, and AI familiarity. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Table B.13: AI and Label Effects on Manipulation and Correct Recognition (Logit)

	Manipulation (<i>AI = Yes</i>)				OriginCorrect (= <i>Yes</i>)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Label Control	-0.005 (0.113)	0.008 (0.114)			0.033 (0.113)	0.028 (0.114)		
Easy AI	2.05*** (0.127)	2.11*** (0.130)	2.05*** (0.127)	2.11*** (0.130)	0.919*** (0.125)	0.914*** (0.126)	0.919*** (0.125)	0.914*** (0.126)
Easy AI + Label	2.44*** (0.138)	2.49*** (0.140)			1.32*** (0.137)	1.33*** (0.137)		
Hard AI	0.310*** (0.110)	0.308*** (0.111)	0.310*** (0.110)	0.308*** (0.111)	-0.803*** (0.111)	-0.803*** (0.112)	-0.803*** (0.111)	-0.803*** (0.112)
Hard AI + Label	0.644*** (0.110)	0.666*** (0.111)			-0.459*** (0.110)	-0.475*** (0.111)		
Label			-0.005 (0.113)	0.008 (0.114)			0.033 (0.113)	0.028 (0.114)
Easy AI × Label			0.388** (0.187)	0.371** (0.189)			0.373** (0.185)	0.389** (0.187)
Hard AI × Label			0.339** (0.156)	0.350** (0.158)			0.310** (0.156)	0.300* (0.158)
Pseudo R ²	0.146	0.157	0.146	0.157	0.088	0.096	0.088	0.096
Observations	4,196	4,196	4,196	4,196	4,196	4,196	4,196	4,196
Image FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	Yes	No	Yes	No	Yes	No	Yes
Mean DV	0.561	0.561	0.561	0.561	0.652	0.652	0.652	0.652

Notes: Each column reports logit estimates. *Manipulation* equals 1 if respondents identified the image as AI-generated, 0 otherwise. *OriginCorrect* equals 1 if respondents correctly recognized whether the image was AI-generated or not, 0 otherwise. *Easy AI* equals 1 for treatment arms Easy AI Unlabeled (EAI) and Easy AI Labeled (LEAI), 0 otherwise; *Hard AI* equals 1 for Hard AI Unlabeled (HAI) and Hard AI Labeled (LHAI), 0 otherwise; *Label* equals 1 if a label was present (Label Control, LEAI, LHAI), 0 otherwise. Controls include age, gender, education, income, political orientation, newsletter subscription, and AI familiarity. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Table B.14: AI and Label Effects on Requesting the Newsletter

	Demand (Newsletter = Yes)											
	(1) OLS	(2) Logit	(3) OLS	(4) Logit	(5) OLS	(6) Logit	(7) OLS	(8) Logit	(9) OLS	(10) Logit	(11) OLS	(12) Logit
Label Control	-0.014 (0.025)	-0.061 (0.112)	-0.020 (0.025)	-0.093 (0.115)	-0.004 (0.033)	-0.019 (0.138)	-0.010 (0.032)	-0.045 (0.140)	-0.032 (0.040)	-0.159 (0.198)	-0.038 (0.039)	-0.202 (0.203)
Easy AI	0.018 (0.026)	0.076 (0.111)	0.019 (0.025)	0.088 (0.115)	0.096* (0.050)	0.395** (0.201)	0.104** (0.049)	0.450** (0.210)	0.195*** (0.052)	0.835*** (0.222)	0.215*** (0.051)	0.967*** (0.231)
Easy AI + Label	-0.024 (0.025)	-0.106 (0.113)	-0.025 (0.025)	-0.120 (0.116)	0.066 (0.057)	0.270 (0.231)	0.048 (0.057)	0.204 (0.244)	0.164*** (0.059)	0.709*** (0.249)	0.159*** (0.059)	0.721*** (0.263)
Hard AI	-0.006 (0.026)	-0.028 (0.112)	-0.006 (0.025)	-0.031 (0.115)	-0.009 (0.034)	-0.040 (0.143)	-0.018 (0.033)	-0.085 (0.147)	0.089** (0.038)	0.400** (0.173)	0.092** (0.037)	0.432** (0.178)
Hard AI + Label	-0.021 (0.025)	-0.093 (0.113)	-0.016 (0.025)	-0.074 (0.115)	-0.090*** (0.034)	-0.398*** (0.154)	-0.086*** (0.034)	-0.405** (0.158)	0.009 (0.038)	0.041 (0.182)	0.025 (0.037)	0.112 (0.187)
Manipulation					-0.098*** (0.037)	-0.439*** (0.168)	-0.111*** (0.036)	-0.517*** (0.173)				
Label Control × Manipulation					-0.027 (0.051)	-0.140 (0.242)	-0.028 (0.050)	-0.158 (0.247)				
Easy AI × Manipulation					-0.043 (0.060)	-0.149 (0.259)	-0.042 (0.060)	-0.158 (0.270)				
Easy AI + Label × Manipulation					-0.047 (0.066)	-0.181 (0.282)	-0.021 (0.066)	-0.076 (0.296)				
Hard AI × Manipulation					0.023 (0.052)	0.106 (0.234)	0.046 (0.051)	0.214 (0.241)				
Hard AI + Label × Manipulation					0.161*** (0.051)	0.723*** (0.233)	0.169*** (0.050)	0.794*** (0.239)				
OriginCorrect									0.098*** (0.037)	0.439*** (0.168)	0.111*** (0.036)	0.517*** (0.173)
Label Control × OriginCorrect									0.027 (0.051)	0.140 (0.242)	0.028 (0.050)	0.158 (0.247)
Easy AI × OriginCorrect									-0.240*** (0.060)	-1.03*** (0.258)	-0.264*** (0.060)	-1.19*** (0.268)
Easy AI + Label × OriginCorrect									-0.244*** (0.066)	-1.06*** (0.282)	-0.243*** (0.066)	-1.11*** (0.296)
Hard AI × OriginCorrect									-0.174*** (0.052)	-0.773*** (0.235)	-0.176*** (0.051)	-0.820*** (0.243)
Hard AI + Label × OriginCorrect									-0.035 (0.051)	-0.156 (0.234)	-0.053 (0.050)	-0.240 (0.240)
Pseudo R ²	0.002	0.002	0.038	0.038	0.010	0.010	0.046	0.047	0.010	0.010	0.046	0.047
Observations	4,196	4,196	4,196	4,196	4,196	4,196	4,196	4,196	4,196	4,196	4,196	4,196
Image FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	No	Yes	Yes	No	No	Yes	Yes	No	No	Yes	Yes
Mean DV	0.349	0.349	0.349	0.349	0.349	0.349	0.349	0.349	0.349	0.349	0.349	0.349

Notes: *Demand* equals 1 if respondents said they wish to receive the Newsletter, 0 otherwise. *Manipulation* equals 1 if respondents identified the image as AI-generated, 0 otherwise. *OriginCorrect* equals 1 if respondents correctly recognized whether the image was AI-generated or not, 0 otherwise. Control is the omitted treatment category in all columns. Even-numbered columns report logit estimates; odd-numbered columns report OLS estimates. Controls include age, gender, education, income, political orientation, newsletter subscription, and AI familiarity. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Table B.15: Mechanisms: Effects of AI and Label on Perceived Outlet Quality

Model:	z-index (1)	Not Accurate (2)	Not Trustworthy (3)	Low Quality (4)	Biased (5)	Very Complex (6)	Not Entertaining (7)	No Researcher Trust (8)
Label Control	0.057 (0.076)	-0.007 (0.037)	-0.011 (0.038)	0.009 (0.037)	0.013 (0.043)	0.052* (0.030)	-0.081* (0.043)	0.015 (0.032)
Easy AI	0.681*** (0.094)	-0.172*** (0.035)	-0.170*** (0.037)	-0.158*** (0.035)	-0.045 (0.051)	0.024 (0.036)	-0.180*** (0.051)	-0.124*** (0.028)
Easy AI + Label	0.499*** (0.082)	-0.202*** (0.032)	-0.214*** (0.032)	-0.130*** (0.040)	-0.084 (0.057)	0.033 (0.041)	-0.246*** (0.058)	-0.142*** (0.025)
Hard AI	0.622*** (0.069)	-0.188*** (0.029)	-0.182*** (0.030)	-0.159*** (0.029)	-0.071* (0.038)	0.024 (0.026)	-0.085** (0.038)	-0.114*** (0.025)
Hard AI + Label	0.659*** (0.070)	-0.202*** (0.029)	-0.205*** (0.029)	-0.163*** (0.030)	-0.087** (0.040)	0.053* (0.028)	-0.125*** (0.041)	-0.135*** (0.024)
OriginCorrect	0.649*** (0.067)	-0.189*** (0.028)	-0.191*** (0.029)	-0.164*** (0.029)	-0.083** (0.038)	0.023 (0.025)	-0.170*** (0.038)	-0.122*** (0.024)
Label Control × OriginCorrect	-0.107 (0.093)	0.009 (0.040)	0.010 (0.041)	0.009 (0.041)	0.025 (0.054)	-0.033 (0.038)	0.092* (0.054)	-0.004 (0.035)
Easy AI × OriginCorrect	-1.20*** (0.108)	0.314*** (0.040)	0.320*** (0.042)	0.343*** (0.041)	0.089 (0.060)	-0.006 (0.042)	0.224*** (0.060)	0.223*** (0.033)
Easy AI + Label × OriginCorrect	-0.957*** (0.098)	0.353*** (0.038)	0.351*** (0.038)	0.267*** (0.044)	0.124* (0.064)	-0.042 (0.046)	0.264*** (0.065)	0.240*** (0.030)
Hard AI × OriginCorrect	-1.29*** (0.095)	0.392*** (0.040)	0.402*** (0.041)	0.357*** (0.041)	0.213*** (0.053)	-0.041 (0.036)	0.165*** (0.053)	0.273*** (0.035)
Hard AI + Label × OriginCorrect	-1.22*** (0.093)	0.376*** (0.038)	0.365*** (0.039)	0.350*** (0.040)	0.156*** (0.053)	-0.052 (0.037)	0.218*** (0.053)	0.237*** (0.032)
R ²	0.115	0.074	0.073	0.077	0.064	0.040	0.055	0.046
Observations	4,196	4,196	4,196	4,196	4,196	4,196	4,196	4,196
Image FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean DV	0	0.146	0.156	0.171	0.422	0.144	0.517	0.1

Notes: Each column reports OLS estimates of treatment effects on perceived outlet quality measures. *z-index* is a standardized composite index averaging *NotAccurate*, *NotTrustworthy*, and *LowQuality* (higher = worse perceived quality). *NotAccurate*, *NotTrustworthy*, and *LowQuality* are binary indicators equal to 1 if the respondent selected one of the two lowest categories on the respective 1–5 Likert scale, 0 otherwise. *Biased*, *VeryComplex*, *NotEntertaining*, and *NoResearcherTrust* are similarly coded binary indicators. *OriginCorrect* equals 1 if respondents correctly recognized whether the image was AI-generated or not, 0 otherwise. Control is the omitted treatment category. Controls include age, gender, education, income, political orientation, newsletter subscription, and AI familiarity. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Table B.16: AI and Label Effects on Requesting the Newsletter and Perceived Outlet Quality (IV estimates)

	Demand (<i>Newsletter = Yes</i>)				z-index			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Manipulation	0.007 (0.036)	0.010 (0.036)			-0.366*** (0.066)	-0.351*** (0.066)		
OriginCorrect			0.015 (0.050)	0.009 (0.049)			-0.348*** (0.091)	-0.345*** (0.091)
Observations	4,196	4,196	4,196	4,196	4,196	4,196	4,196	4,196
F-test (1st stage), Manipulation	171.1	174.1			171.1	174.1		
F-test (1st stage), OriginCorrect			91.2	90.3			91.2	90.3
Wald (2nd stage)	0.033	0.084	0.089	0.037	30.5	28.6	14.8	14.5
Image FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	Yes	No	Yes	No	Yes	No	Yes
Mean DV	0.349	0.349	0.349	0.349	0	0	0	0

Notes: Each column reports 2SLS estimates where *Manipulation* or *OriginCorrect* is instrumented by treatment assignment. *Demand* equals 1 if respondents said they wish to receive the Newsletter, 0 otherwise. *z-index* is a standardized composite index of perceived outlet quality (higher = better). *Manipulation* equals 1 if respondents identified the image as AI-generated, 0 otherwise. *OriginCorrect* equals 1 if respondents correctly recognized whether the image was AI-generated or not, 0 otherwise. Even-numbered columns include controls for age, gender, education, income, political orientation, newsletter subscription, and AI familiarity. All columns include image fixed effects. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Table B.17: Mechanism: AI and Label Effects on Perceived Outlet Quality (IV estimates)

	z-index (1)	Not Accurate (2)	Not Trustworthy (3)	Low Quality (4)	Biased (5)	Very Complex (6)	Not Entertaining (7)	No Researcher Trust (8)
OriginCorrect	-0.345*** (0.091)	0.108*** (0.038)	0.084** (0.039)	0.126*** (0.039)	-0.046 (0.050)	0.0005 (0.036)	-0.166*** (0.052)	0.053 (0.032)
Observations	4,196	4,196	4,196	4,196	4,196	4,196	4,196	4,196
F-test (1st stage), OriginCorrect	90.3	90.3	90.3	90.3	90.3	90.3	90.3	90.3
Wald (2nd stage)	14.5	8.20	4.74	10.1	0.836	0.0002	10.2	2.68
Image FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean DV	0	0.146	0.156	0.171	0.422	0.144	0.517	0.1

Notes: Each column reports 2SLS estimates where *OriginCorrect* is instrumented by treatment assignment (Easy AI, Hard AI, Label, and their interactions). *z-index* is a standardized composite index of perceived outlet quality (higher = better). *NotAccurate*, *NotTrustworthy*, and *LowQuality* are binary indicators equal to 1 if the respondent selected one of the two lowest categories on the respective 1–5 Likert scale, 0 otherwise. *Biased*, *VeryComplex*, *NotEntertaining*, and *NoResearcherTrust* are similarly coded binary indicators. *OriginCorrect* equals 1 if respondents correctly recognized whether the image was AI-generated or not, 0 otherwise. Controls include age, gender, education, income, political orientation, newsletter subscription, and AI familiarity. All columns include image fixed effects. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Table B.18: Journalists' Sample Characteristics

Characteristic	Category	N	Value
Age	Mean (SD)	132	35.1 (10.4)
Gender	Female	79	59.8%
Gender	Male	50	37.9%
Gender	Non-binary	2	1.5%
Gender	Prefer not to say	1	0.8%
Country (top 10)	United States of America	40	35.1%
Country (top 10)	United Kingdom	24	21.1%
Country (top 10)	South Africa	12	10.5%
Country (top 10)	Germany	8	7%
Country (top 10)	Mexico	8	7%
Country (top 10)	Spain	7	6.1%
Country (top 10)	Canada	4	3.5%
Country (top 10)	Italy	4	3.5%
Country (top 10)	Poland	4	3.5%
Country (top 10)	France	3	2.6%
Employment status	Working full time now	94	71.2%
Employment status	Working part time now	27	20.5%
Employment status	Student	6	4.5%
Employment status	Unemployed	3	2.3%
Employment status	Temporarily laid off	1	0.8%
Employment status	Other	1	0.8%
Currently a journalist	Yes	101	76.5%
Currently a journalist	No, but I have in the past	26	19.7%
Currently a journalist	No, I have never worked as journalist	5	3.8%
Outlet type	Online Newspaper	38	29.2%
Outlet type	Freelance	34	26.2%
Outlet type	TV	20	15.4%
Outlet type	Daily Newspaper	16	12.3%
Outlet type	Weekly Newspaper	14	10.8%
Outlet type	Blog	8	6.2%
Tenure in journalism	More than 5 years	60	83.3%
Tenure in journalism	Less than 1 year	12	16.7%
Role	Editor	54	41.5%
Role	Reporter	51	39.2%
Role	Opinion Writer	18	13.8%
Role	Columnist	7	5.4%

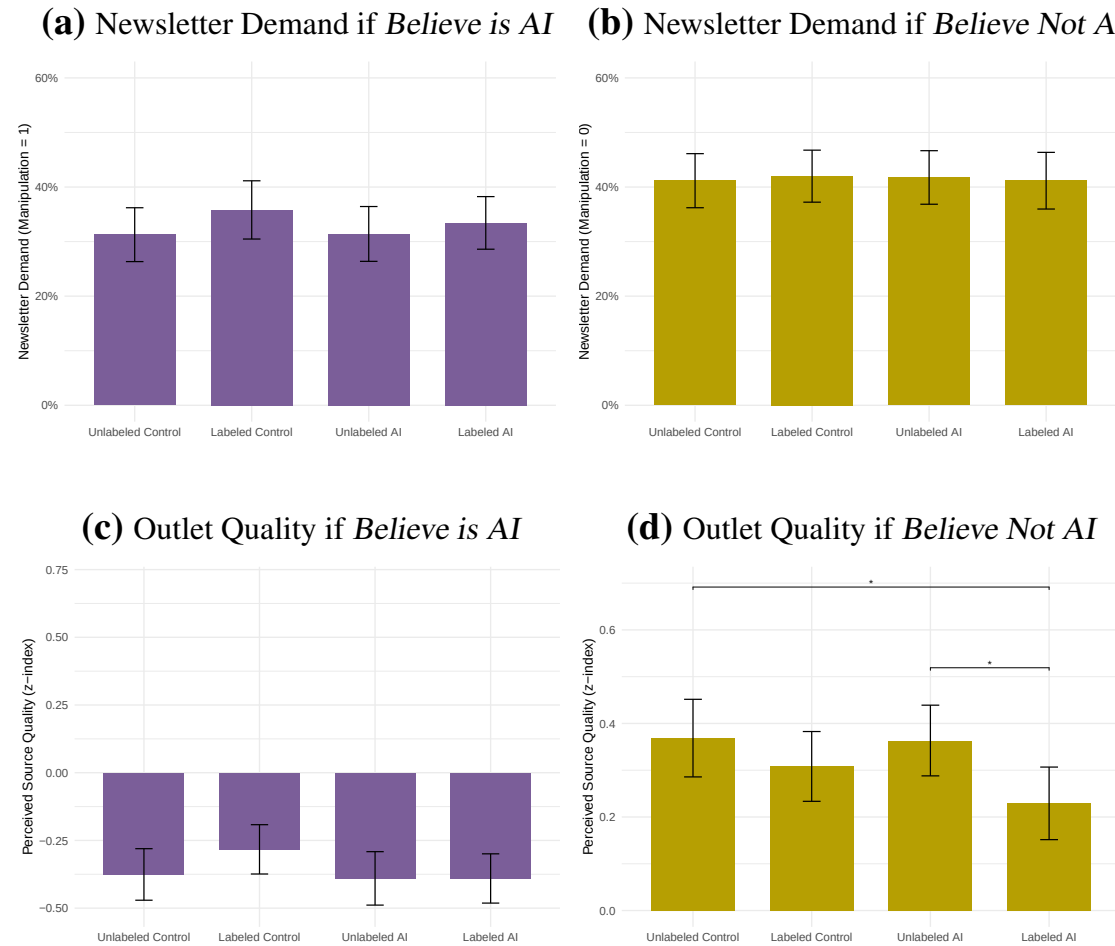
Table B.19: Journalists' Reported Reasons for Using AI in Their Work

Topic Category	% of Respondents	Illustrative Example
Efficiency and Time Pressure	35.7%	<i>"Journalists are using AI to reduce the time consumption it takes to research, gather and present their work as they have tight deadlines to meet."</i>
Research, Data, and Information Processing	31%	<i>"AI can be used as a tool by subeditors to generate quick alt copy for images (for visually impaired readers) or by reporters to collate data."</i>
Professional Concerns and Ethical Issues	29.4%	<i>"Any journalist who uses AI in their articles is a disgrace to their profession. It's lazy, insulting and morally wrong."</i>
Workflow Facilitation and Content Scaling	27%	<i>"AI has come to facilitate journalistic work, and in an environment where a very high workload is constantly demanded, AI is a tool that allows journalists to meet their bosses' demands."</i>
Image and Content Generation	22.2%	<i>"Journalists use it to generate pictures which are sometimes hard to find and AI can assist with creating those images."</i>
Writing and Editing Support	22.2%	<i>"I think journalists use AI to make their articles more intriguing. They make their articles to seem more in depth and interesting."</i>
Transparency, Training, and Ethical Frameworks	12.7%	<i>"I think journalists are using AI in their articles as a way to speed up the writing process. When working to tight deadlines, using AI can be a way to ensure accuracy in spelling and grammar, which will result in less time spent proof reading."</i>

Notes: Categories are not mutually exclusive; respondents could mention multiple themes. Percentages reflect the share of the 126 respondents whose answer contained at least one reference to the given theme.

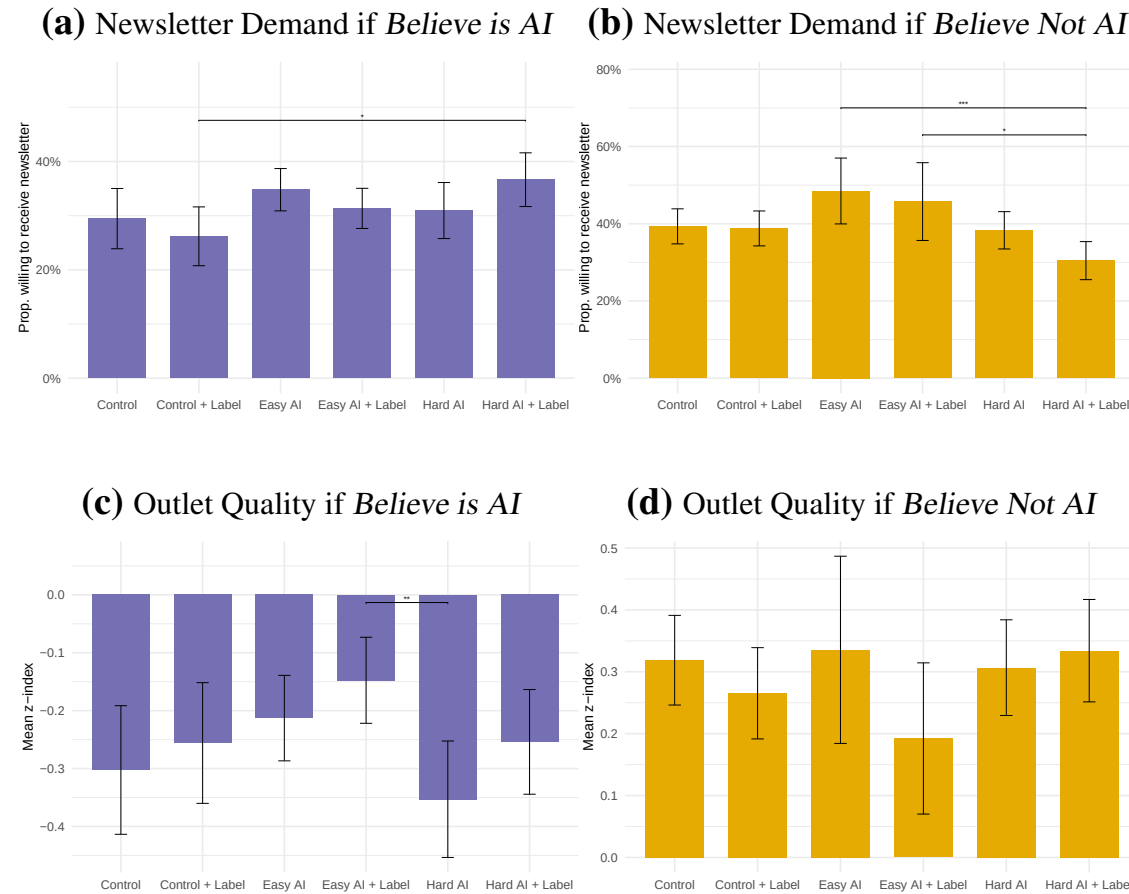
C Appendix Figures

Figure C.1: Newsletter Demand and Perceived Outlet Quality by Treatment and AI Beliefs



Notes: Panel (a and b) show the share of respondents demanding the newsletter by treatment arm. Panel (c and d) show the average perceived outlet quality by treatment arm. Perceived outlet quality is a composite index averaging standardized responses to three 1–5 scale items: *Accuracy*, *Trustworthiness*, and *Quality*. Panels (a and c) and (b and d) split the sample by whether respondents believed the image was AI-generated (*Manipulation* = 1) or not (*Manipulation* = 0) respectively. Error bars represent 95% confidence intervals; stars above brackets denote significance levels (** $p < 0.01$, * $p < 0.05$, * $p < 0.10$).

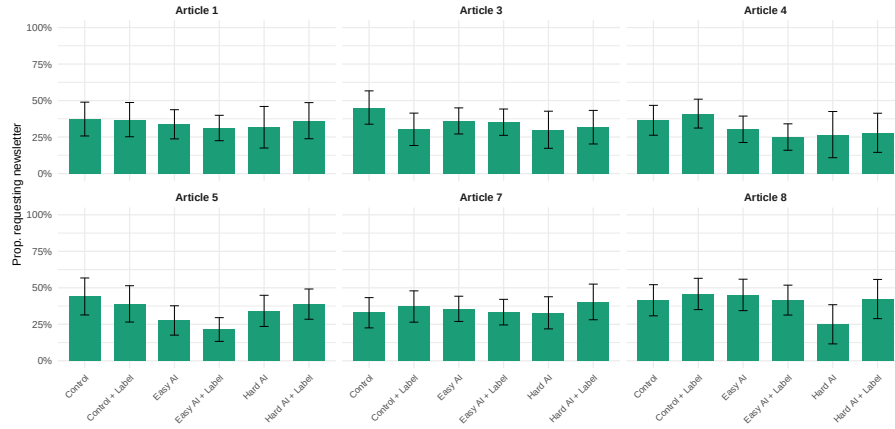
Figure C.2: Newsletter Demand and Perceived Outlet Quality by Treatment and AI Beliefs



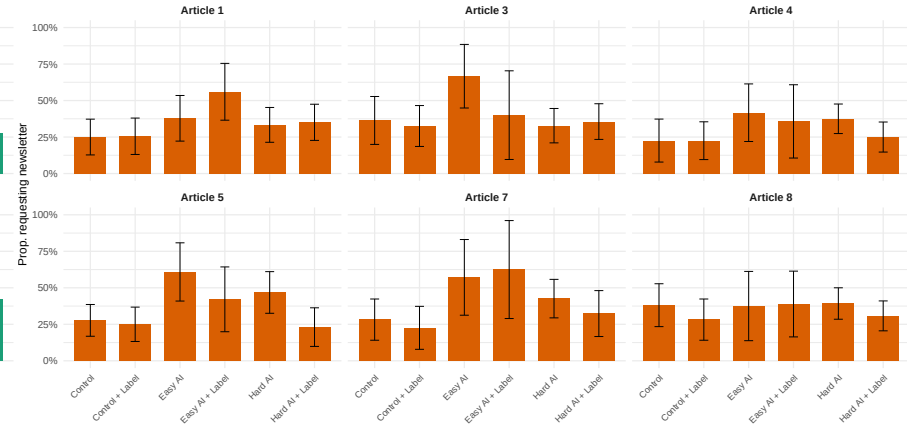
Notes: Panel (a and b) show the share of respondents demanding the newsletter by treatment arm. Panel (c and d) show the average perceived outlet quality by treatment arm. Perceived outlet quality is a composite index averaging standardized responses to three 1–5 scale items: *Accuracy*, *Trustworthiness*, and *Quality*. Panels (a and c) and (b and d) split the sample by whether respondents believed the image was AI-generated (*Manipulation* = 1) or not (*Manipulation* = 0) respectively. Error bars represent 95% confidence intervals; stars above brackets denote significance levels (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$).

Figure C.3: Newsletter Demand by Treatment and Article

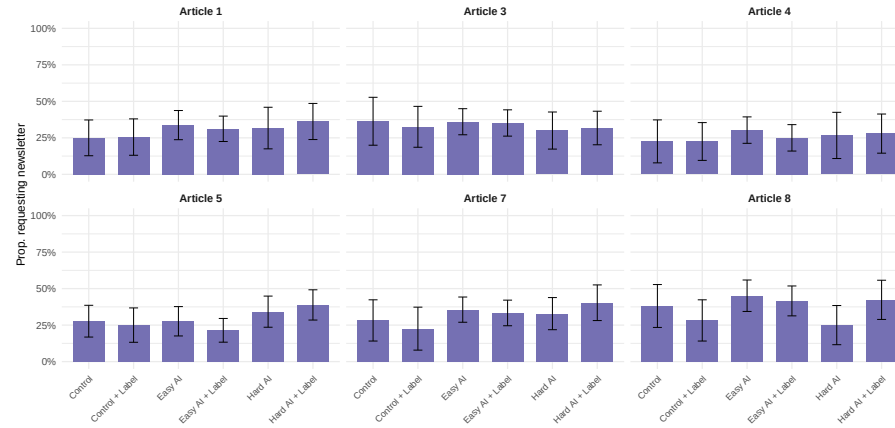
(a) Newsletter Demand if *Correct AI/Not*



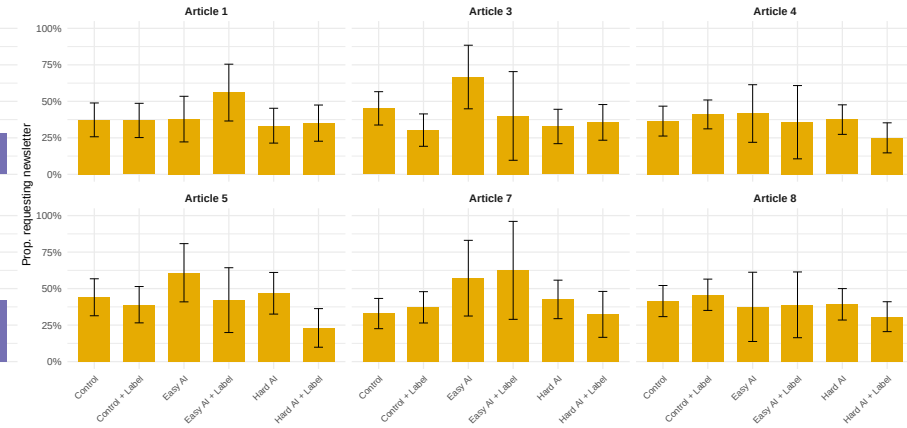
(b) Newsletter Demand if *Incorrect AI/Not*



(c) Newsletter Demand if *Believe is AI*

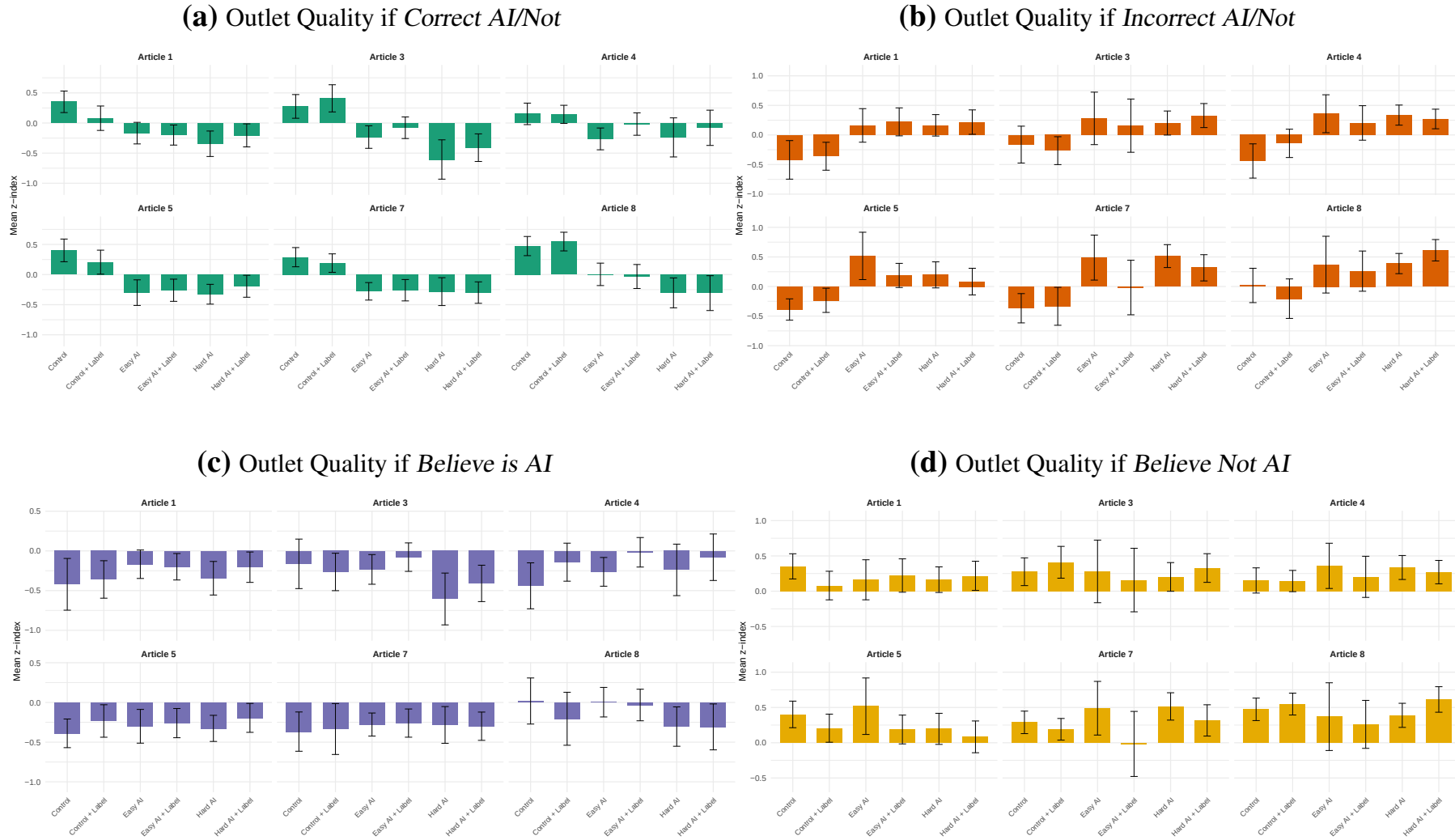


(d) Newsletter Demand if *Believe Not AI*



Notes: Each panel corresponds to one of the six articles shown to respondents. Panels (a)–(b) and (c)–(d) split the sample by whether respondents correctly identified the origin of the image (*OriginCorrect* = 1 and = 0 respectively) or by whether they believed it was AI-generated (*Manipulation* = 1 and = 0 respectively). Bar heights show the share of respondents requesting the newsletter within each treatment arm and article. Error bars represent 95% confidence intervals. Pairwise significance tests are not reported as the within-article sample size is insufficient to power them reliably.

Figure C.4: Perceived Outlet Quality by Treatment and Article



Notes: Each panel corresponds to one of the six articles shown to respondents. Panels (a)–(b) and (c)–(d) split the sample by whether respondents correctly identified the origin of the image (*OriginCorrect* = 1 and = 0 respectively) or by whether they believed it was AI-generated (*Manipulation* = 1 and = 0 respectively). Bar heights show the share of respondents requesting the mean z-index within each treatment arm and article. Error bars represent 95% confidence intervals. Pairwise significance tests are not reported as the within-article sample size is insufficient to power them reliably.

D Image Generation Prompts

Experiment 1 The AI image was generated using DALL·E. The following prompt was used to produce a photorealistic version closely matching the original wire-service photograph:

“Please create an AI version of the original image, photorealistic, shot on Canon EOS R5, 85mm lens, f/1.8 aperture, 8K resolution, professional photography, hyper-detailed skin texture, volumetric lighting, HDR.”

Experiment 2 Image generation followed a two-step protocol designed to separate scene description from image generation. In the first step, a dedicated instance of ChatGPT was provided with the original article page and instructed to produce a detailed, literal description of the accompanying photograph — covering objects, setting, composition, perspective, lighting, and salient visual elements — without any reference to provenance, realism, or artistic style.

In the second step, a separate instance of ChatGPT Image (with no access to the original photograph) was provided with the textual description from step one and instructed to generate an image matching the described scene and composition. At this stage, photorealism was varied across two versions: one prompted to be highly photorealistic (“very realistic AI”), closely resembling a professional news photograph, and one prompted to be intentionally less photorealistic (“not very realistic AI”), with visual cues making AI generation more apparent. The depicted scene, composition, and semantic content were held constant across versions.

E Instructions

A full print preview of the Qualtrics survey for Experiment 1 (including all instructions and blocks) is available at the following [link](#).

The instructions of Experiment 2 are nearly identical to those of Experiment 1, but with two additional treatments. The complete set of instructions and article pictures can be found in the following [link](#).